CIRCULAR CLASSIFIER

A MINI PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

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BONAFIDE CERTIFICATE

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Declaration

DECLARATION

We hereby declare that the project report entitled "CIRCULAR CLASSIFIER" submitted to the Department of Computer Applications, Thiagarajar College of Engineering, Madurai, is a record of the original work carried out by our team. All sources of information and contributions of others that were referred to during this project have been duly acknowledged with proper references. This work has not been submitted earlier, either in part or in full, for the award of any other degree or diploma of this or any other institution.

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Abstract

In today's fast-paced academic environment, higher education institutions issue a large number of circulars, notices and announcements every day covering examinations, placements, academic schedules, holidays and administrative instructions. These circulars are usually distributed through email groups, notice boards and messaging applications, making important information difficult to track, search and prioritise. As a result, students and staff often miss critical deadlines. The Circular Classifier has therefore been developed as a centralized institutional circular management system that automatically collects incoming circulars, extracts their text, classifies them into meaningful categories, detects expiry dates and organises everything by priority for easy retrieval.

The system has been designed and implemented as a full-stack web application using React with Vite (TypeScript and Tailwind CSS) for the front-end and Node.js with Express for the back-end. Circulars are received from a mail server through IMAP, processed by background workers built on Redis and Bull queues, and analysed using Tesseract OCR together with PDF and DOCX text extraction. A keyword classifier automatically assigns each circular to a category such as Examinations, Academic, Placements, Holidays, Events or Administration, while an expiry-date detector highlights the circulars that are ending soon. All records, categories, users and logs are stored in MongoDB, and the application provides a dashboard, instant keyword search with filters and an administrative console.

Deploying this system offers several advantages to the institution. It removes the dependence on scattered email threads and notice boards, ensures that every circular is captured and archived in one place, and greatly reduces the effort required to find an old notice. Priority-based sorting and expiry alerts help students and staff act on time-critical announcements, while the classification and search modules make large volumes of information easy to navigate. The modular architecture allows new categories and notification channels to be added without disturbing existing modules. In the long term, the system can be extended with mobile notifications, multi-institution support and analytics dashboards.

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LIST OF ABBREVIATIONS

S.No	Abbreviation	Full Form
1	HTML	HyperText Markup Language
2	CSS	Cascading Style Sheets
3	JS	JavaScript
4	PaaS	Platform as a Service
5	UI	User Interface
6	UX	User Experience
7	API	Application Programming Interface
8	DFD	Data Flow Diagram
9	UML	Unified Modeling Language
10	IDE	Integrated Development Environment
11	ISO	International Organization for Standardization
12	VS Code	Visual Studio Code
13	Git	Global Information Tracker
14	B2B	Business to Business
15	SEO	Search Engine Optimization

1. INTRODUCTION

1.1 INTRODUCTION

React.js

The front-end of the Circular Classifier is built with React, a JavaScript library developed by Meta for creating dynamic and reusable user-interface components. Each part of the application, such as the dashboard, the circular table, the search panel and the admin console, is implemented as an independent, reusable component. React's virtual DOM updates only the parts of a page that change, keeping large circular listings fast and responsive, and its declarative style makes the code predictable and easy to debug while improving maintainability and scalability.

JavaScript (ES6)

JavaScript, particularly the ES6 (ECMAScript 2015) version, is used throughout the client to add interactivity and logical control. Features such as arrow functions, template literals, promises and modular imports keep the code concise and maintainable. JavaScript handles user events, asynchronous API calls, live search and the dynamic rendering of circular data, ensuring that the interface responds instantly to user actions.

GitHub for Collaboration

GitHub is used for version control and collaborative development. The team stored the source code in GitHub repositories, created branches for individual features and tracked every change through commits and pull requests. This produced a clear version history, minimised conflicts between the two developers and allowed features to be integrated smoothly throughout the project.

Cloud Deployment (PaaS)

The application is deployed on a cloud Platform-as-a-Service environment so that it remains accessible to all users without dedicated institutional servers. Continuous deployment from the GitHub repository keeps the live build up to date, and a global content-delivery network reduces latency for users. The same deployment pipeline serves the Node.js API and the static client build, providing a simple and reliable hosting
setup.

1.2 ORGANISATION PROFILE

Thiagarajar College of Engineering (TCE), Madurai, established in 1957, is a government-aided autonomous institution affiliated to Anna University, Chennai. It is one of the most reputed engineering colleges in Tamil Nadu, offering undergraduate, postgraduate and doctoral programmes in engineering, technology, architecture and computer applications, and is accredited by the National Assessment and Accreditation Council
(NAAC).

The Department of Computer Applications offers the Master of Computer Applications (MCA) programme and focuses on software engineering, web technologies, databases and emerging areas such as data science and cloud computing. The department provides well-equipped laboratories, high-speed internet and a project-oriented curriculum that encourages students to build complete, deployable software systems under the guidance of experienced faculty.

The vision of the department is to produce competent and socially responsible computer professionals who can design and deliver innovative software solutions. Its mission is to impart quality technical education, promote research and teamwork, and nurture ethical values. These objectives guided the development of the Circular Classifier as a practical, socially useful project that addresses a genuine communication problem faced by institutions.

The strengths of the department lie in its experienced faculty, well-structured curriculum and active project culture. Students are encouraged to work in teams, adopt modern development tools and follow standard software processes, which closely matches the way the Circular Classifier project was planned, executed and documented by the team.

In line with the department's emphasis on full-stack and cloud-ready skills, the project uses a modern technology stack that includes React with Vite and TypeScript, Node.js with Express, MongoDB, Redis and Bull background workers, Tesseract OCR and a keyword-based classifier. The institution actively promotes such end-to-end projects, as they prepare students for real-world software development and provide useful tools for the campus community. With its strong academic foundation, technical infrastructure and supportive mentoring, the department provided an ideal environment in which the Circular Classifier was designed, implemented and evaluated.

1.3 EXISTING SYSTEM

Module description	Existing manual method used	Drawbacks of the existing method	Features proposed to overcome
Distribution	Emails, notice boards and messaging groups	Circulars are scattered across mails and easily missed	Automatic ingestion from email and uploads into one searchable and central repository
Classification	Manual tagging or no labelling	Notices are hard to find by topic	Keyword-based classifier assigns categories automatically
Search	Scrolling through long mail threads	Old circulars are difficult to retrieve	Instant keyword search with category, priority, tag and
Priority & Expiry management	No concept of priority or deadlines	Urgent notices and deadlines are missed, urgent mails remain unnoticed, and priority is	Automatic priority levels and expiry-date alerts with reminders for pending actions
Dashboard	No single view of active circulars	Users cannot see what needs attention or which	Dashboard sorted by date and priority with an 'Ending Today' view
Attachments	Files attached to separate emails and scattered across	Attachments lose context and are hard to locate	Attachments stored and linked to their circulars in one place
Users	Open mailing lists and chat groups	No control over who publishes notices or edits them after publishing, and no audit trail	Role-based accounts for students, staff and administrators with per-role publish, edit and delete

2. PROBLEM DESCRIPTION

2.1 PROBLEM STATEMENT

Educational institutions publish a constant stream of circulars covering examinations, results, placements, fee deadlines, holidays and administrative instructions. In most institutions these circulars are shared through email groups, printed notice boards or messaging applications, and there is no single place where they are stored or searched. Important announcements get buried under newer messages, duplicate copies circulate in different groups, and students frequently miss deadlines. Tracking which circulars are still valid, which are urgent and which relate to a particular department is extremely difficult.

The absence of a centralized platform also creates extra work for the administrative staff, who must resend the same information repeatedly and answer the same queries. Hence there is a clear need for a professional, searchable and priority-aware circular management system that can serve as a single source of truth for every institutional announcement.

2.2 OBJECTIVES

- To design and develop a centralized web application that collects institutional circulars from email and file sources automatically.
- To extract text from PDF, DOCX and image attachments using Tesseract OCR and document-parsing libraries.
- To classify circulars into categories with a keyword-based classifier and to assign priority levels automatically.
- To detect expiry dates and highlight circulars that are ending soon through dashboards and alerts.
- To provide instant keyword search with category, priority and date filters.
- To build an administrative console for managing categories, keywords and user accounts, and to use GitHub for collaboration and version control.

3. SYSTEM REQUIREMENT SPECIFICATION

3.1 SCOPE

This project focuses on developing a centralized circular management system for educational institutions. It collects circulars from a mail server and from uploaded documents, processes them automatically, classifies them by category and presents them through a searchable, priority-aware web interface.

The scope covers the complete workflow of circular handling: ingestion, text extraction, classification, expiry detection, storage, search and administration. It is designed for institutional use but can be adapted to any organization that distributes large volumes of notices.

The system is built as a full-stack web application, with a React (Vite, TypeScript and Tailwind CSS) client, a Node.js and Express REST API, MongoDB for storage, and Redis with Bull for background jobs. Tesseract OCR, PDF and DOCX parsers are used for text extraction. It is version-controlled with GitHub and deployed as a Node.js service for reliable, always-available access.

3.2 FUNCTIONAL REQUIREMENTS

- The system should automatically fetch incoming circulars from the configured mail server and also support manual document upload.
- It should extract text from email bodies and from PDF, DOCX and image attachments, including scanned documents through OCR.
- It should classify each circular into a category using configurable keywords and assign an appropriate priority.
- It should detect expiry dates and clearly indicate circulars that are expiring today or soon.
- It should provide instant keyword search with filters for category, priority, tag, sender and date range.
- It should present circulars on a dashboard sorted by date and priority, with a detailed circular view.
- It should provide an admin console to manage categories, keywords and user accounts, and must remain responsive across devices.

3.2.1 SOFTWARE SPECIFICATION

Frontend Framework – React.js

The client of the Circular Classifier is built using React, a JavaScript library for building interactive, component-based user interfaces. Each major part of the application, such as the dashboard, the circular table, the search panel and the admin console, is implemented as a reusable component. This modular approach keeps the interface consistent, simplifies maintenance and allows new screens to be added easily, while React's virtual DOM keeps large circular listings fast and responsive.

Styling – Tailwind CSS

The interface is styled using Tailwind CSS, a utility-first CSS framework that provides precise control over layout, typography, spacing and colour. Tailwind keeps the look and feel consistent across every screen and device, and its responsive utilities make the dashboard, tables and forms adapt cleanly to desktops, tablets and mobile phones.

Backend Framework – Node.js and Express

The back-end is built with Node.js and the Express framework. It exposes a REST API for authentication, circulars, categories, search and user management. Redis and Bull queues process heavy jobs such as extraction, classification and expiry detection asynchronously, keeping the interface responsive.

Development Environment – Visual Studio Code (VS Code)

The project is developed in Visual Studio Code, a lightweight and feature-rich IDE that supports debugging, extensions, integrated terminals and version control. VS Code provides a flexible environment for writing React components and Node.js modules, managing project files and testing both the client and the server efficiently.

Version Control – Git & GitHub

Git is used to track code changes, while GitHub enables collaborative development by allowing both team members to work on different features simultaneously. It ensures a structured workflow with version history, branch management and pull requests, making teamwork efficient and reducing code conflicts.

Database – MongoDB

All application data, including circulars, users, categories and processing logs, is stored in MongoDB, a document-oriented NoSQL database. MongoDB's flexible schema suits circular data, which varies in structure and attachments, and the Mongoose ODM provides clear models, validation and efficient queries for search and filtering.

3.2.2 HARDWARE SPECIFICATION

Development Environment

For development purposes, a system with an Intel i3 processor or higher, 4 GB RAM (8 GB recommended) and at least 20 GB of free storage is sufficient. This configuration supports running Visual Studio Code, the Node.js server, MongoDB, Redis and a browser simultaneously. A display with 1366 × 768 resolution or higher allows efficient testing of the dashboard and responsive layouts.

Deployment Environment

The application is deployed on a cloud Node.js hosting environment with a managed MongoDB database and a Redis instance for background queues. The platform automatically manages server resources and scaling, ensuring that circulars are processed reliably and the service stays accessible with low latency.

End-User Devices

The application is optimized for a wide range of devices. Users can access it seamlessly from desktops, laptops, tablets and mobile phones, and the responsive design adapts to various screen sizes and resolutions. Administrative users who upload or manage circulars use the same interface, and only an internet connection and a modern browser are required.

4. PROPOSED SYSTEM

4.1 Methodology

The proposed system is a full-stack, service-oriented web application for managing institutional circulars, with a strong emphasis on automation, modular design and usability. The methodology combines a layered client-server architecture with an Agile development process, enabling iterative progress, continuous feedback and flexible enhancements. The approach is described below by feature:

1. Layered Client-Server Architecture
The system is divided into a React client, an Express REST API, background processors and a MongoDB data layer. Each layer has a clear responsibility and communicates through well-defined interfaces, so the client, server and workers can be developed, tested and scaled independently while remaining easy to maintain.

2. Responsive Design
The interface is designed to work seamlessly across desktops, laptops, tablets and mobile devices. Tailwind's responsive utilities adapt the dashboard, tables and forms to different screen sizes, ensuring a consistent and intuitive experience for both staff and students on any device.

3. Asynchronous Processing with Queues
Background jobs are processed asynchronously using Redis and Bull queues. When a circular arrives from the mail server or is uploaded, a job is enqueued for text extraction, OCR, classification and expiry detection. This keeps the API responsive, allows long-running tasks to complete reliably and makes the processing pipeline easy to extend with new steps.

4. Agile Development Methodology
The project follows Agile methodology, incorporating iterative sprints to achieve steady progress. Sprint planning defined tasks for ingestion, extraction, classification, search and the admin console. Each sprint concluded with internal reviews and testing, allowing early detection of errors, continuous improvement and flexibility for design changes.

5. Prototyping and Design Validation
Wireframes and mock-ups were created during the early stages of development to visualise the dashboard, circular table, search and filter panels and admin screens. These prototypes were validated against real circular data and user needs, ensuring that the final interface was clear, consistent and aligned with the project goals.

6. Version Control and Collaboration
Git and GitHub were used for version control and collaborative development. This ensured organised code management, easy rollback to previous versions when needed, and efficient teamwork between the two developers who worked on different modules.

7. Deployment and Accessibility
The application is deployed on a cloud platform that serves the static client build and runs the Node.js API with managed MongoDB and Redis services. This ensures the system is highly available, processes circulars reliably in the background and can be accessed by users across different regions.

4.2 Advantages of the Proposed System

The proposed system provides several distinct advantages, both from a technical and an institutional perspective:

1. Modular and Maintainable Code
The layered architecture ensures that each module is independent and reusable. This improves maintainability, reduces development time for future enhancements, and simplifies debugging and testing.

2. Enhanced User Experience
A responsive, searchable interface with priority sorting and expiry alerts lets students and staff find the right circular quickly, reducing missed deadlines and improving communication across the institution.

3. Iterative Development and Flexibility
Agile methodology allows continuous progress through sprints. Iterative development enables early detection of issues, timely feedback and the flexibility to implement changes without affecting the entire project.

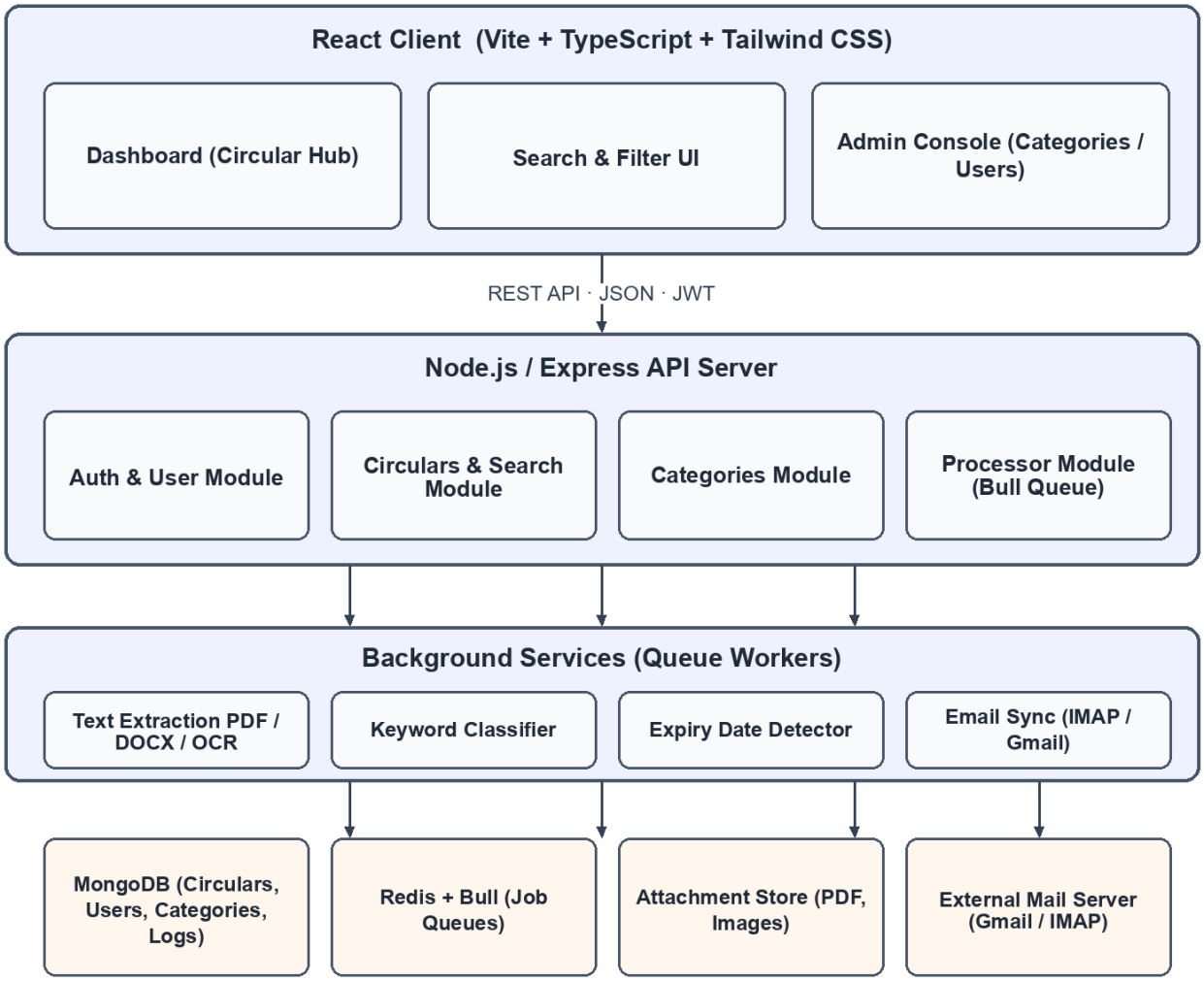
4. Efficient Collaboration
Using Git and GitHub for version control ensures organised teamwork, minimises code conflicts and provides a historical record of changes. This streamlines development in a multi-developer environment.

5. Quick Deployment and Scalability
Cloud deployment with managed services enables automatic builds, background job processing and scalability without additional server setup, allowing the system to handle an increasing number of circulars as the institution grows.

6. Centralized Institutional Communication
The system becomes a single source of truth for institutional announcements. It captures every circular, classifies it automatically and makes it searchable by keyword, category, priority and date, thereby strengthening communication between the administration, faculty and students.

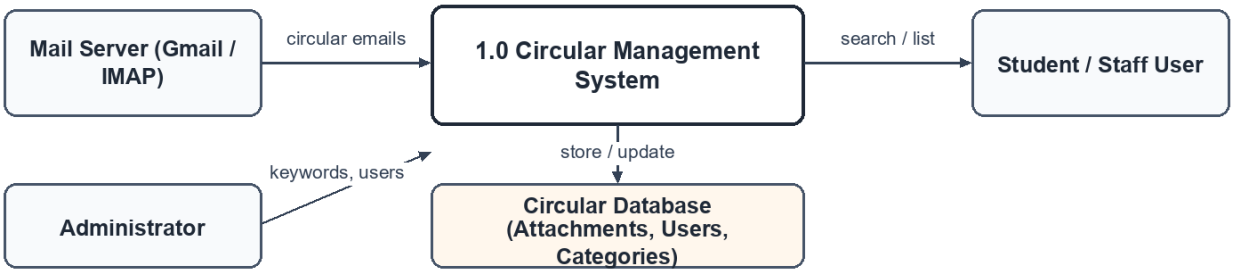
5. SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE DIAGRAM

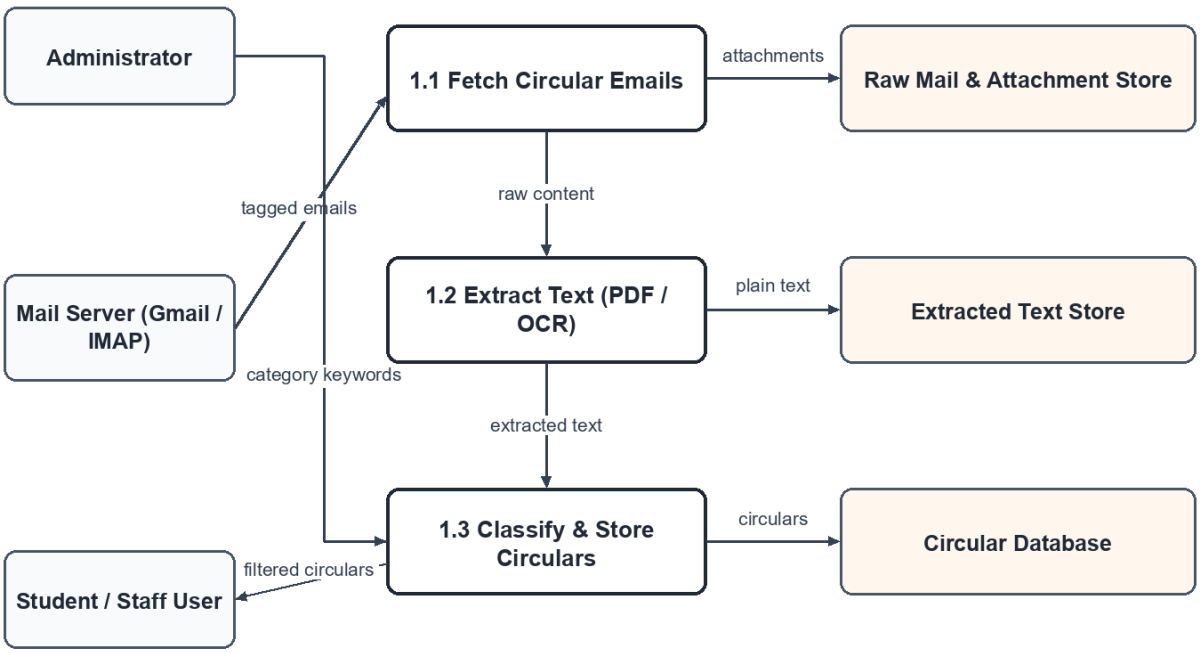


5.2 DATA FLOW DIAGRAM

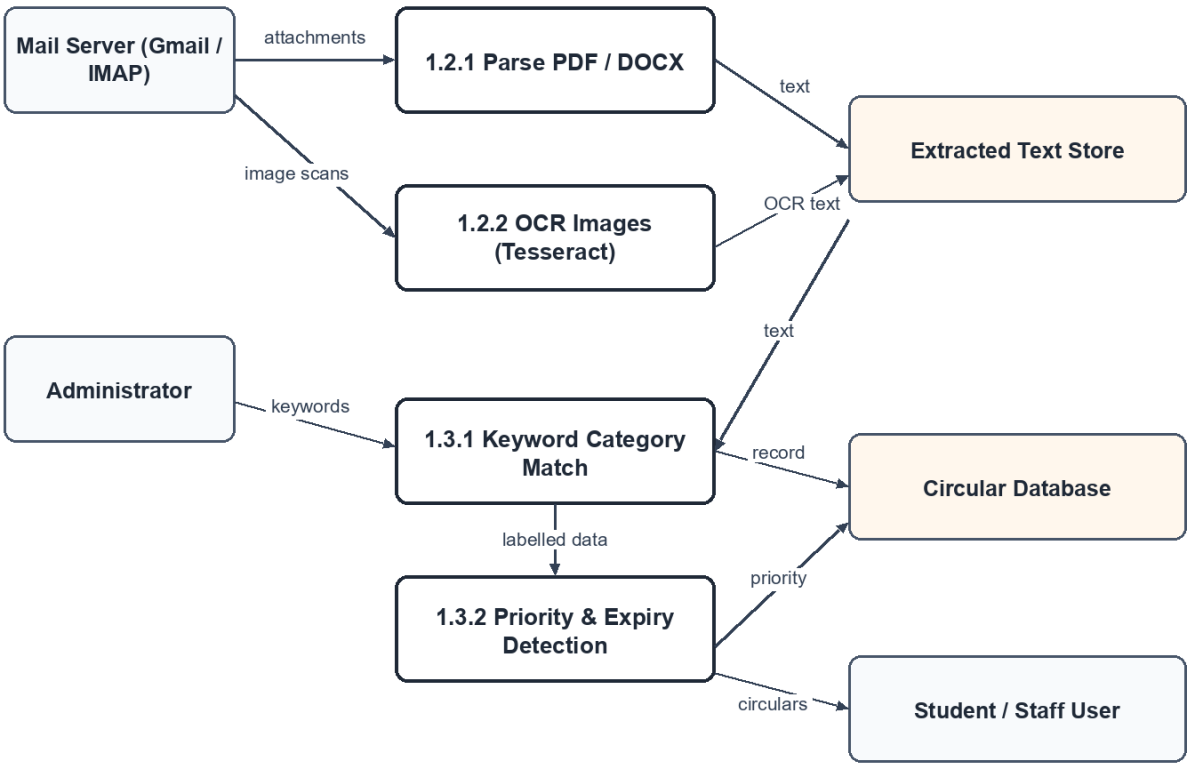
5.2.1 Level 0 DFD



5.2.1 Level 1 DFD

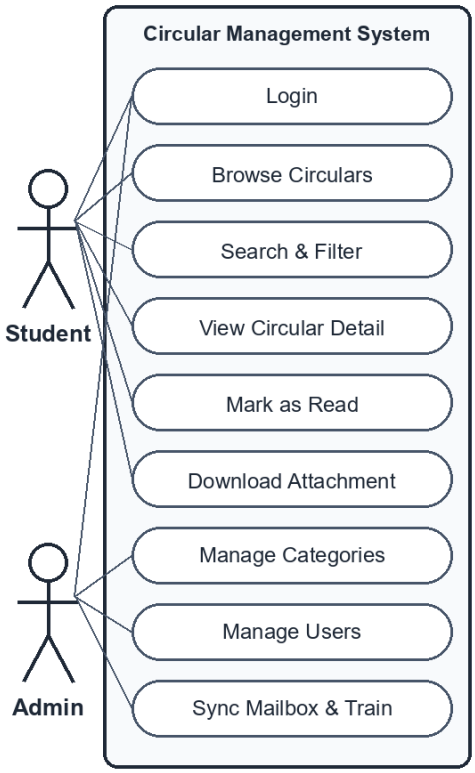


5.2.1 Level 2 DFD



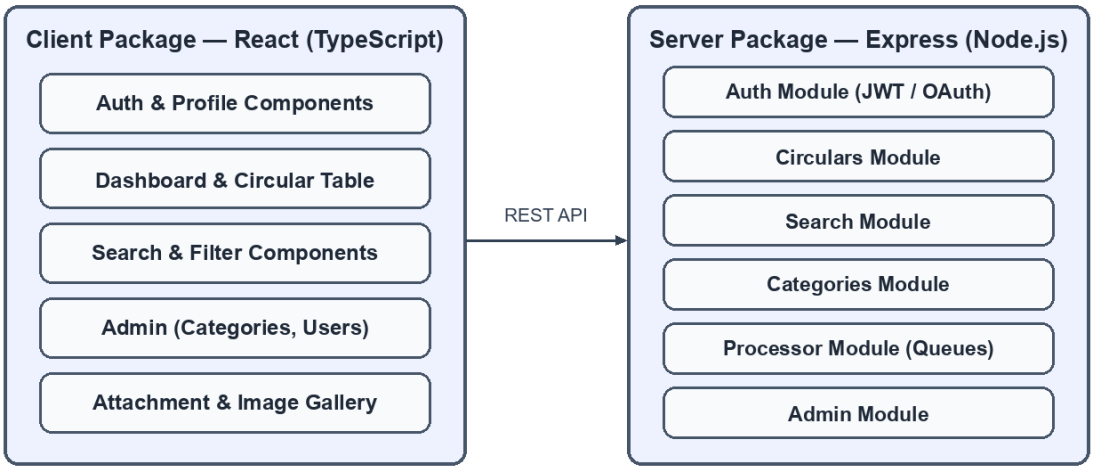
5.3 LOGICAL DESIGN USING UML

5.3.1 Usecase Diagram

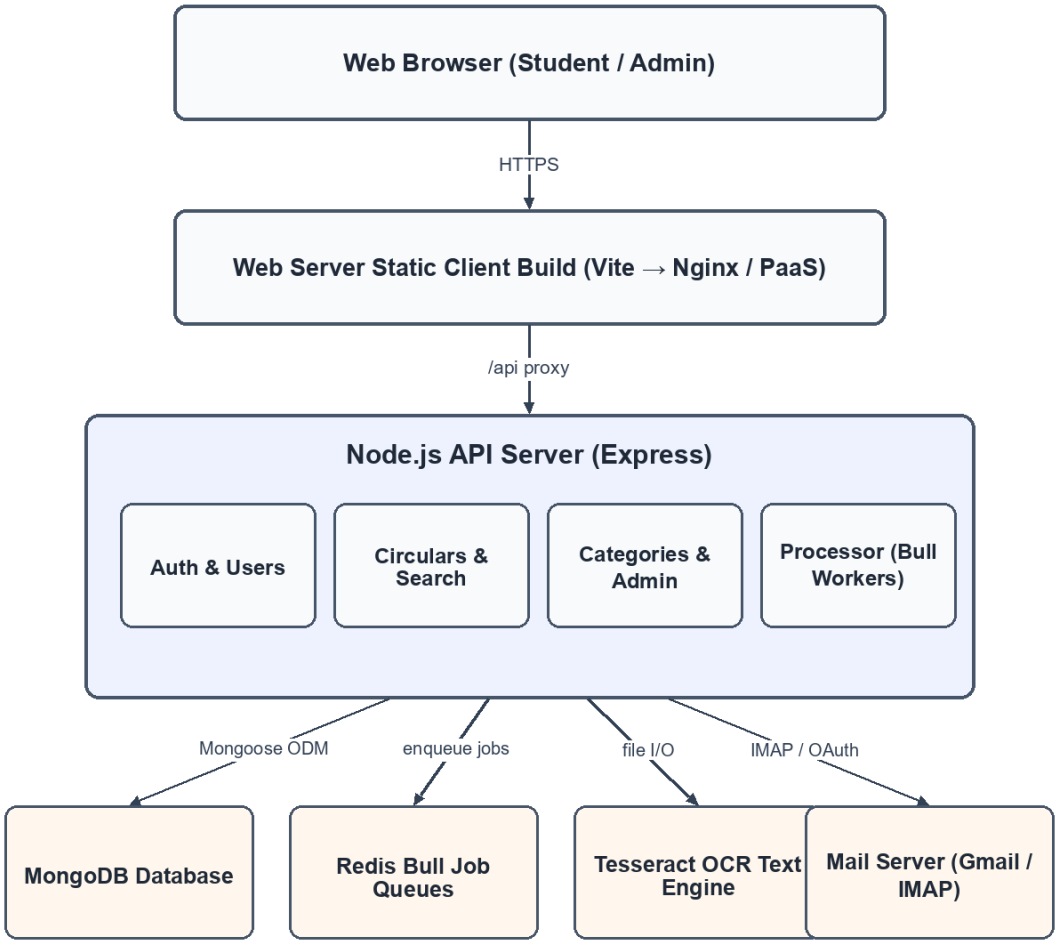


5.4 PHYSICAL DESIGN USING UML

5.4.1 Component Diagram



5.4.2 Deployment Diagram



6. MODULE DESCRIPTION

6.1 Authentication Module

The authentication and user module handles registration, secure login and role-based access. Users register with their name, email, department and account type (Student or Admin), and credentials are protected using hashed passwords and JSON Web Tokens. The module maintains user profiles and permissions, ensuring that administrative functions such as managing categories and users are available only to authorised accounts, while students and staff can view and search
circulars.

6.2 Dashboard Module

The dashboard, presented as the Circular Hub, is the landing screen after login. It lists institutional circulars sorted by date and priority and provides quick access to categories, users and search. A dedicated 'Ending Today' view highlights the circulars whose deadline is the current day, and summary cards show counts such as circulars ending today. Filters for sender, category, priority and date range allow users to narrow the list, making the dashboard a fast, at-a-glance overview of everything that
requires
attention.

6.3 Circulars Module

This module manages the complete life cycle of a circular, from ingestion to display. Circulars received from the mail server or uploaded manually are stored with their title, category, priority, sender, tags, received date and due date. Users can open a detailed view that shows the full content, attachment, extracted important dates and the classification result, and administrators can edit, archive or delete circulars, ensuring that the repository always reflects the current state of institutional communication.

6.4 Email & OCR Module

This module fetches circulars automatically from the configured institutional mail server using IMAP and also accepts manual uploads of PDF, DOCX and image files. Email bodies are parsed, and attachments are processed by PDF and DOCX text extractors. Scanned documents and images are passed through the Tesseract OCR engine to recover their text. Extracted content is normalised and queued for classification and expiry detection, so that even image-based notices become fully
searchable.

6.5 Classification Module

The keyword classification module assigns each circular to a category automatically. Each category, such as Examinations, Academic, Placements, Holidays, Events or Administration, is defined with a set of keywords and phrases. When a circular is processed, the module scores it against these keyword sets and selects the best-matching category, storing the confidence and the matched terms. Administrators can add or edit categories and keywords from the admin console, so classification can be tuned to the institution's own vocabulary without any code changes.

6.6 Expiry Alerts Module

The expiry date detection module scans the extracted text for dates and deadline phrases and derives a due date for each circular. It then computes the number of days remaining and flags circulars that are expiring today or within a short window. These flags drive the 'Ending Today' view and the due-date column on the dashboard, helping users act on time-critical announcements. Validity information is also shown on the detailed circular view so that outdated notices are not mistaken for current ones.

6.7 Search Module

The search and filter module provides instant keyword search across all circulars, with the option to filter by category, priority, tag, sender and date range. Results are ranked and displayed with their category, priority level and received date. This module allows a user to locate a specific circular within seconds even when the repository contains hundreds of notices, combining full-text matching with structured filters for accurate results.

6.8 Admin Module

The admin console gives administrators control over the system configuration and users. It includes category management, where categories and their keywords are created and edited, and user management, where accounts are listed with their role, department and join date. Changes made in the console immediately affect how new circulars are classified, making the console the operational heart of the system.

6.9 Attachments Module

The attachment and storage module manages circular attachments. Attachments such as PDF and image files are stored securely and linked to their circulars, and all records including circulars, users, categories and processing logs are persisted in MongoDB. This module guarantees data integrity and provides the reliable foundation on which search, display and administration depend.

7. RESULTS AND IMPLEMENTATION

RESULTS

The Circular Classifier project delivered a robust, automated platform for managing institutional circulars. The results achieved are as follows:

1. **Interactive and User-Friendly Interface:**

    - The dashboard presents circulars in a clean, sorted layout that is easy to scan and navigate.

    - Users can reach any circular, category or filter within a few clicks, improving overall engagement.

2. **Automated Circular Ingestion:**

    - Circulars are fetched automatically from the configured mail **server and from uploaded documents and attachments.**

    - Circulars are classified into categories with keyword matching and ranked by priority.

3. **Automatic Classification:**

    - Expiry detection flags circulars that are ending today or soon.

4. **Expiry and Priority Alerts:**

    - Priority levels help users focus on the most urgent notices first.

    - Automated processing removes the repeated manual effort of sharing notices.

5. **Instant Search and Filters:**

   - A powerful search lets users find circulars by keyword, category, priority, tag, sender or date.
   - Results are ranked and displayed with their key details for quick decisions.

6. **Responsive and Cross-Device Compatibility:**

   - The application works well on desktops, tablets and mobile devices.
   - Ensures consistent performance and appearance across different screen sizes.

7. **Professional Deployment:**

   - The application is deployed on a cloud platform with managed database and queue services.
   - Provides a secure, reliable and always-available service to the institution.

Overall Outcome: The project successfully achieved its objective of **creating an automated, searchable and priority-aware system** that improves how institutions communicate with students and staff.

Implementation Process

The development of the Circular Classifier was completed in a structured and systematic approach, as described below:

1. **Project Planning:**

   - Defined the scope, objectives and target users.
   - Identified critical features: circular ingestion, extraction, classification, expiry detection and search.
   - Prepared a timeline and allocated tasks to ensure timely completion.

2. **System Design:**

- Designed the navigation structure and data model for intuitive use.
- Planned the user interface and defined the categories and priority rules.

3. **Development:**

- Implemented the client using React with a modular, component-based architecture for maintainability and scalability.
- Styled the interface with Tailwind CSS for a modern, responsive layout.
- Built the Express REST API and background workers for ingestion, OCR, classification and expiry detection.
- Stored circulars, users, categories and logs in MongoDB using clear models.
- Optimized the application for performance and fast loading across devices.

4. **Testing:**

- Functional Testing: Verified ingestion, classification, search and admin features.
- Design Testing: Checked layout consistency and responsiveness across devices and browsers.
- Identified and resolved minor issues before deployment to ensure smooth operation.

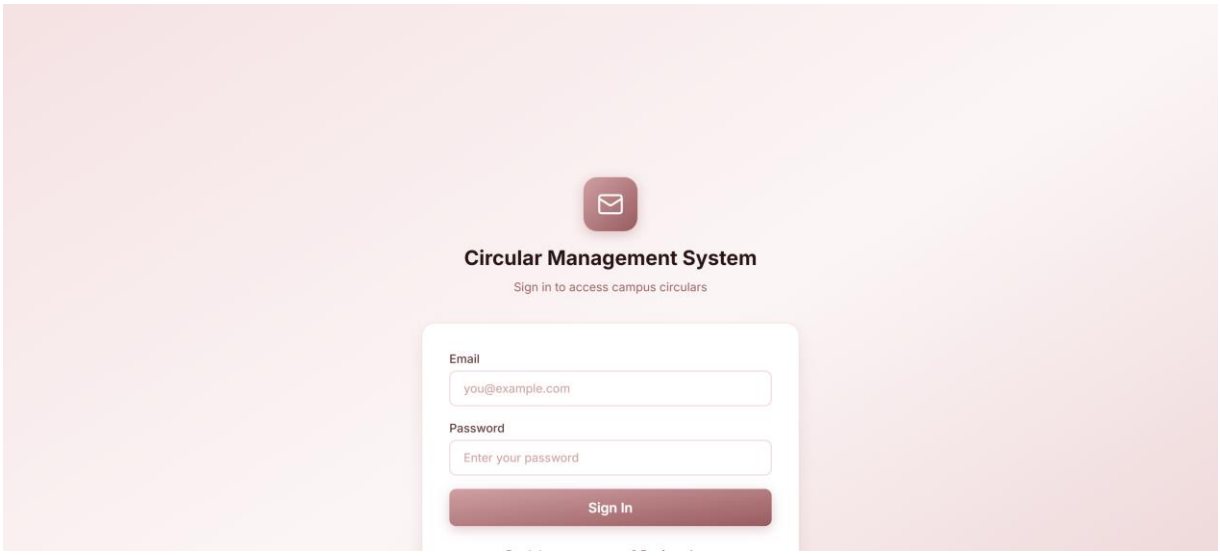
5. **Deployment:**

- Deployed the tested application on a cloud platform with managed services.

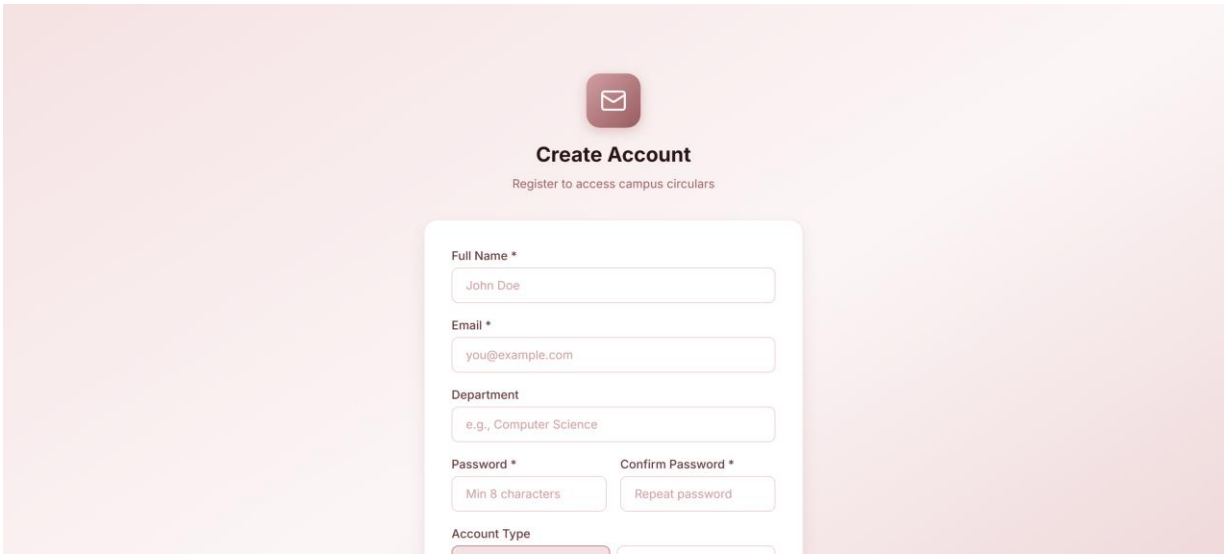
- Configured the environment variables and mail-server settings for the institution.

- Verified live performance, background processing and secure access for users.

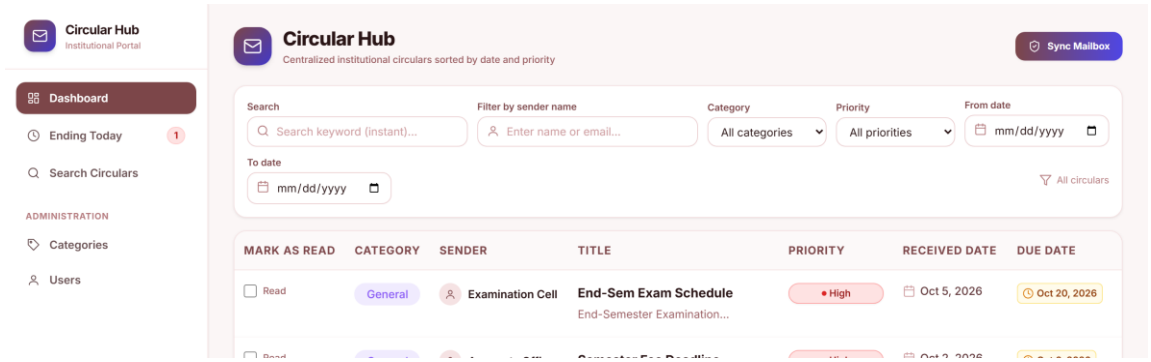
LOGIN PAGE :

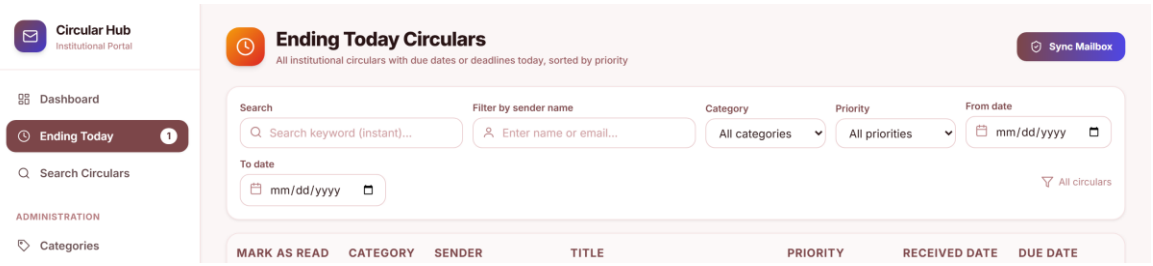


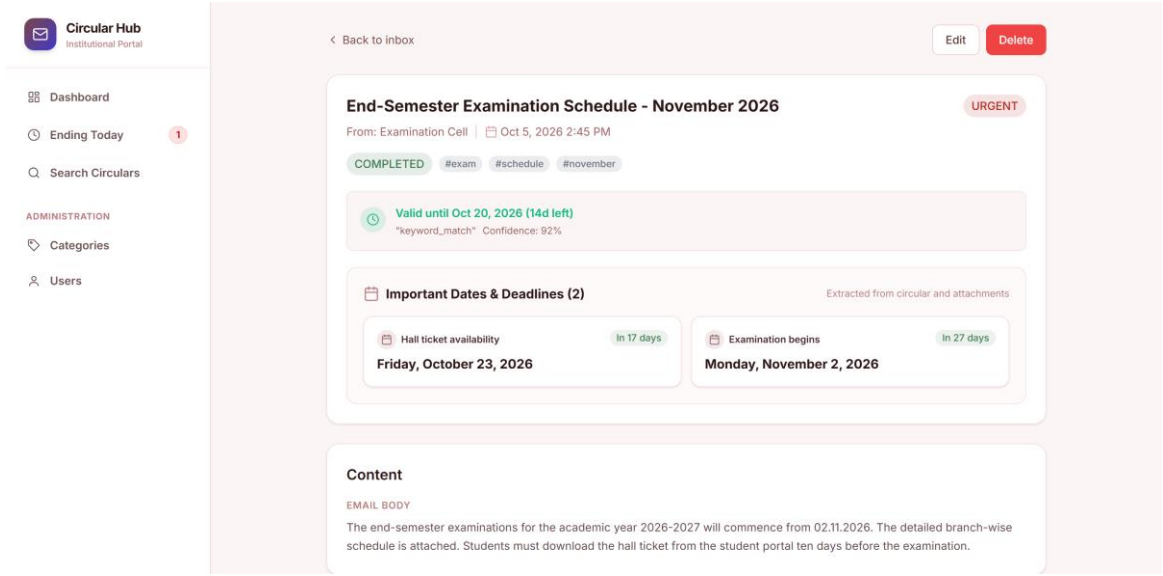
REGISTRATION :



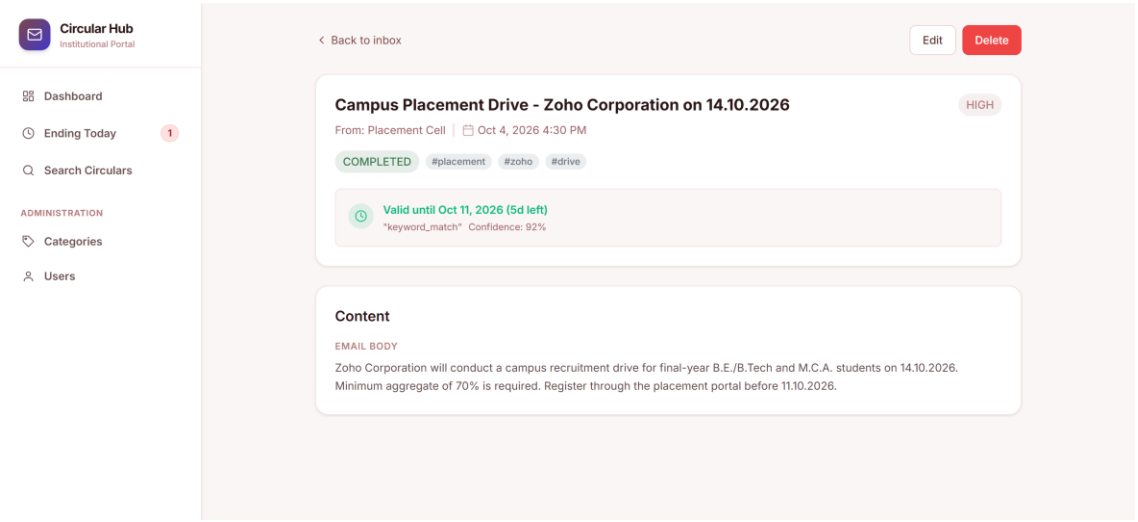
DASHBOARD :


Circular Hub
Institutional Portal
Dashboard
Ending Today
1
Search Circulars
ADMINISTRATION
Categories
Users
Circular Hub
Centralized institutional circulars sorted by date and priority
Sync Mailbox
Search
Search keyword (instant)...
Filter by sender name
Enter name or email...
Category
All categories
Priority
All priorities
From date
mm/dd/yyyy
To date
mm/dd/yyyy
All circulars
MARK AS READ
CATEGORY
SENDER
TITLE
PRIORITY
RECEIVED DATE
DUE DATE
Read
General
Examination Cell
End-Sem Exam Schedule
End-Semester Examination...
High
Oct 5, 2026
Oct 20, 2026

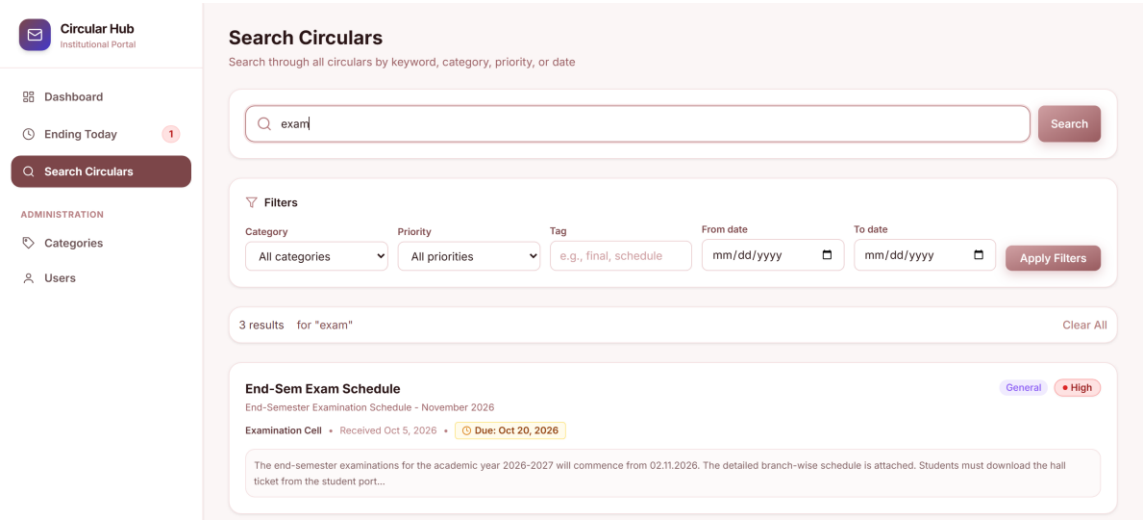

Circular Hub
Institutional Portal
Dashboard
Ending Today
1
Search Circulars
ADMINISTRATION
Categories
Ending Today Circulars
All institutional circulars with due dates or deadlines today, sorted by priority
Sync Mailbox
Search
Search keyword (instant)...
Filter by sender name
Enter name or email...
Category
All categories
Priority
All priorities
From date
mm/dd/yyyy
To date
mm/dd/yyyy
All circulars
MARK AS READ
CATEGORY
SENDER
TITLE
PRIORITY
RECEIVED DATE
DUE DATE

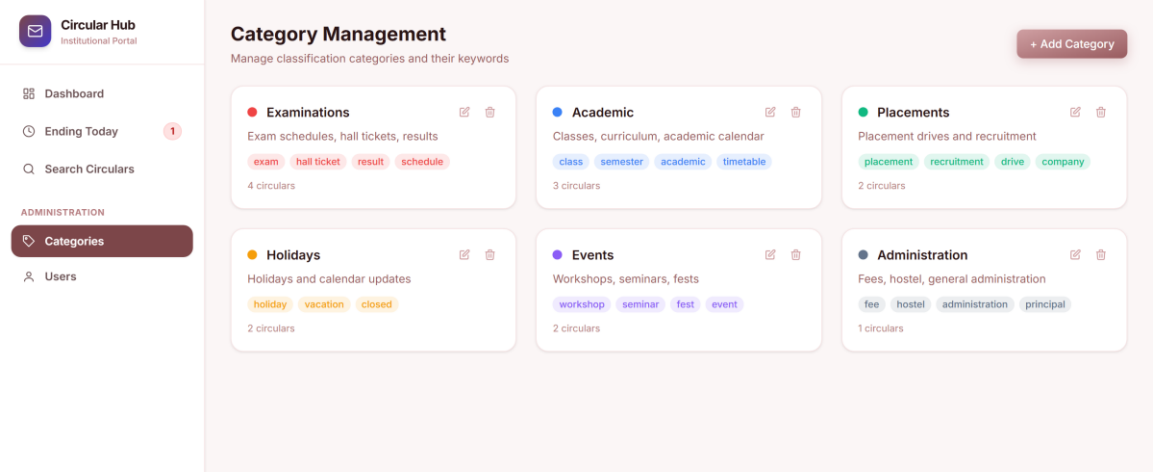

Circular Hub
Institutional Portal
Dashboard
Ending Today
1
Search Circulars
ADMINISTRATION
Categories
Users
Back to inbox
Edit
Delete
End-Semester Examination Schedule - November 2026
URGENT
From: Examination Cell
Oct 5, 2026 2:45 PM
COMPLETED
#exam
#schedule
#november
Valid until Oct 20, 2026 (14d left)
"keyword_match" Confidence: 92%
Important Dates & Deadlines (2)
Extracted from circular and attachments
Hall ticket availability
In 17 days
Friday, October 23, 2026
Examination begins
In 27 days
Monday, November 2, 2026
Content
EMAIL BODY
The end-semester examinations for the academic year 2026-2027 will commence from 02.11.2026. The detailed branch-wise schedule is attached. Students must download the hall ticket from the student portal ten days before the examination.

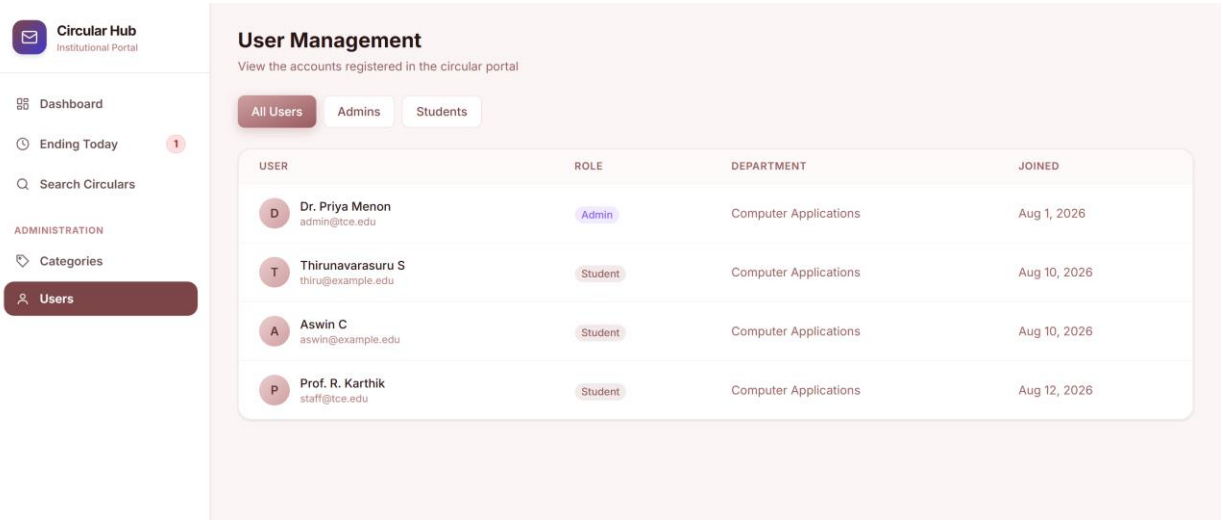
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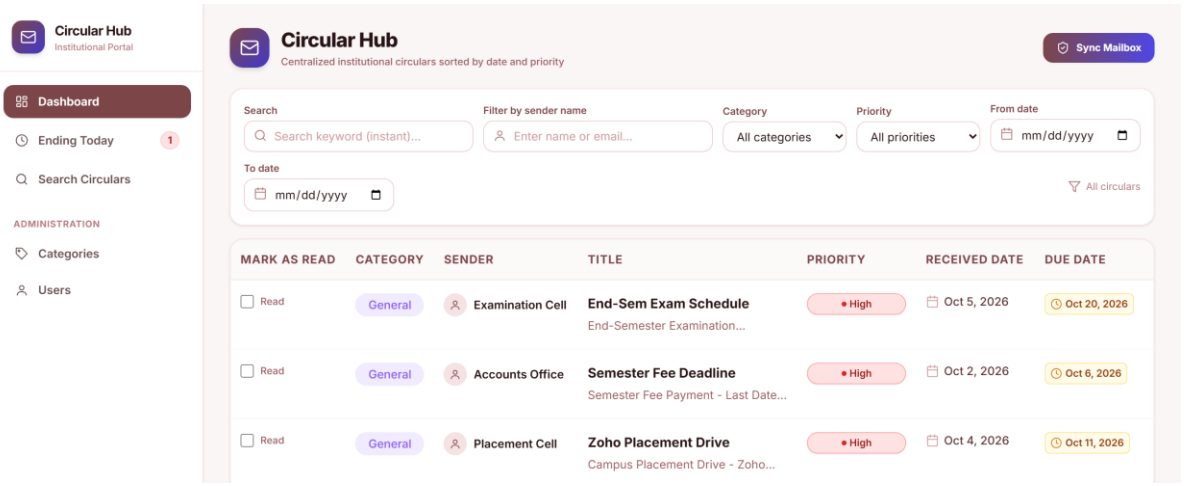


SEARCH & ADMIN :

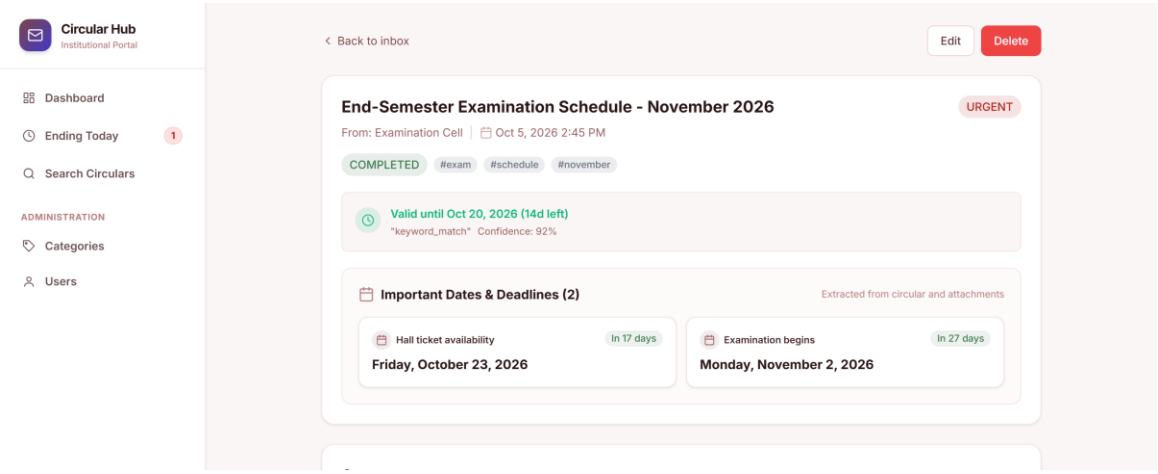


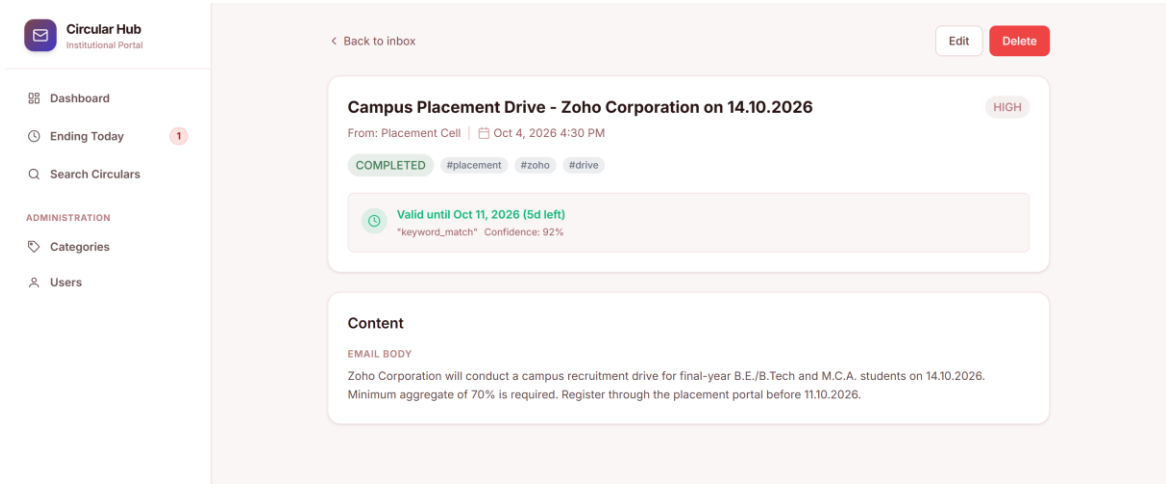


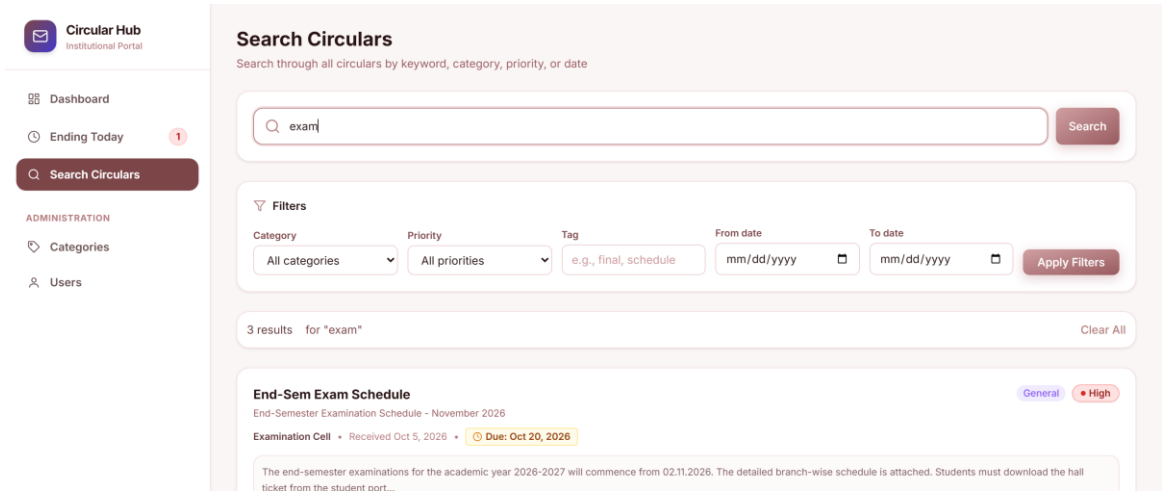




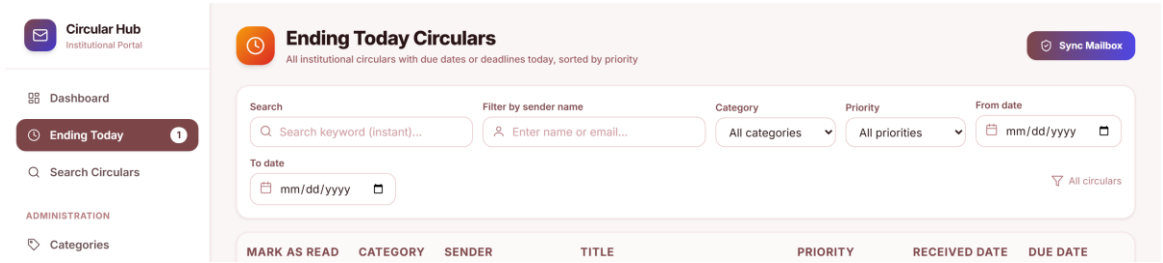
CIRCULAR DETAILS :







EXPIRY ALERTS :



Implementation Summary: Through careful planning, systematic design, rigorous testing and cloud deployment, the Circular Classifier project successfully delivered a robust, reliable and user-centric platform that meets the institution's requirements and improves communication and engagement.

8. CONCLUSION

The Circular Classifier project has been a successful endeavour in delivering an automated, searchable and user-centric platform for institutional circulars. The project achieved all its primary objectives, providing both functional and practical value to students, staff and administrators.

Key points of the conclusion include:

1. **Successful Deployment:**

    ○ The application was successfully deployed on a cloud platform with managed database and queue services.
    professional and always-available service.

    ○ Live deployment ensures that circulars are accessible to users across the institution at any time.

2. User Feedback and Iteration:

    ○ The system was well received by students and staff, who continue to provide feedback for further improvements.

    ○ Continuous feedback allows iterative updates, ensuring the platform evolves according to user needs.

3. **Enhanced User Experience:**

    ○ The dashboard, instant search and filters make it easy to find and track circulars.

    ○ Users can explore circulars, categories and deadlines seamlessly, improving awareness of important announcements.
    announcements.

4. Comprehensive Circular Repository:

- A centralized repository captures circulars from email and uploads and classifies them automatically for easy retrieval.

- Search, filters and expiry alerts help users make informed and timely decisions.

5. **Accessibility & Responsiveness:**

- The platform is fully responsive and optimized for desktops, tablets and mobile devices.

- Accessibility features, such as clear labels and keyboard navigation, ensure a wider reach and inclusivity.

6. **Modern Technology Stack:**

- **Using React, Node.js, MongoDB and Redis allowed for a scalable and maintainable application.**

- The modular architecture enables easy updates, new categories and future scalability.

7. **Testing & Quality Assurance:**

- The project underwent rigorous functional and design testing, ensuring reliable performance and eliminating errors before deployment.

Overall, the Circular Classifier project stands as a successful implementation of modern full-stack development practices, meeting institutional requirements and delivering a platform that is functional, reliable, accessible and scalable. It not only centralizes institutional communication but also provides a strong base for continuous improvement, with user feedback driving future enhancements.

9.BIBILOGRAPHY

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   https://developer.mozilla.org

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   https://www.mongodb.com/docs

5. Redis and Bull Documentation – For background job queues and asynchronous processing. https://docs.bullmq.io

6. Tesseract OCR Documentation – For extracting text from images and scanned documents. https://tesseract-ocr.github.io

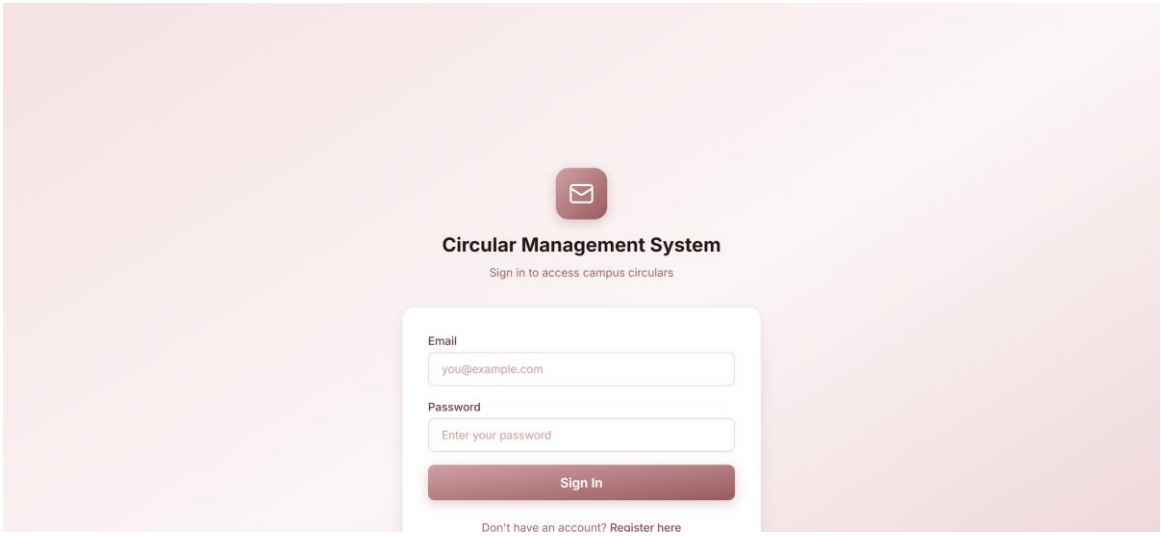
7. Tailwind CSS Documentation – For utility-first styling and responsive design. https://tailwindcss.com/docs

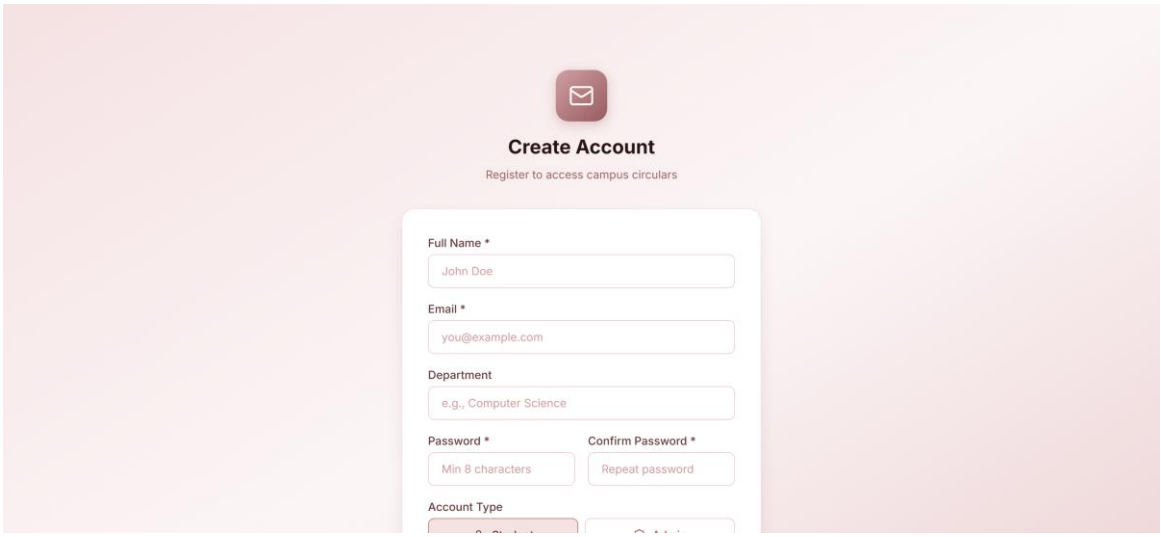
8. Project Requirement Documents – Internal project documentation prepared by the team for the Circular
   Classifier.

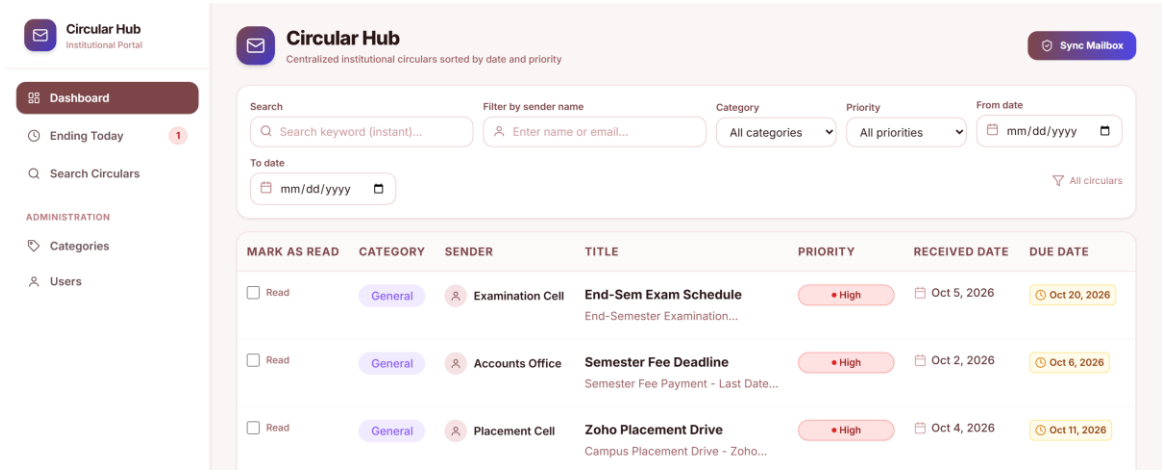
10. APPENDICES

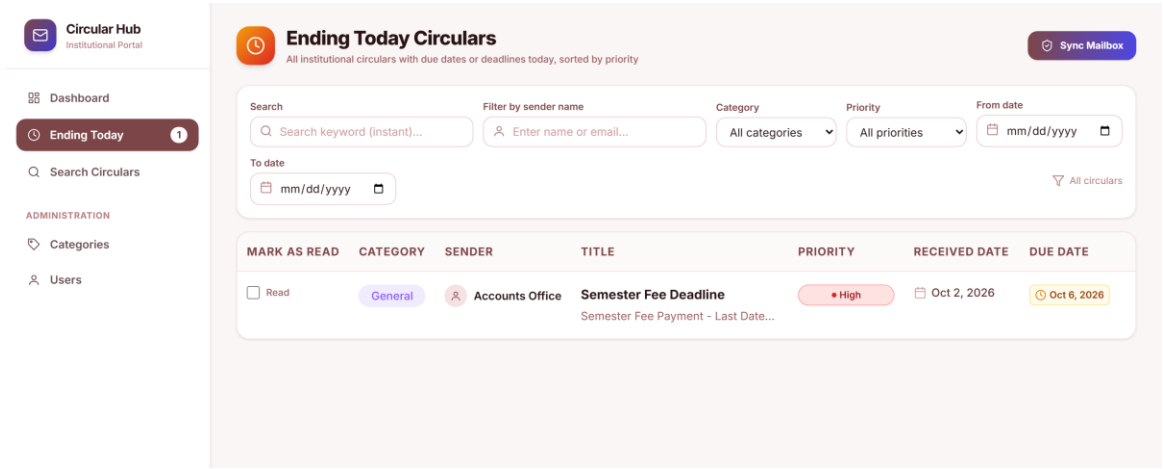
10.1 APPENDIX - A PROJECT SCREENSHOTS

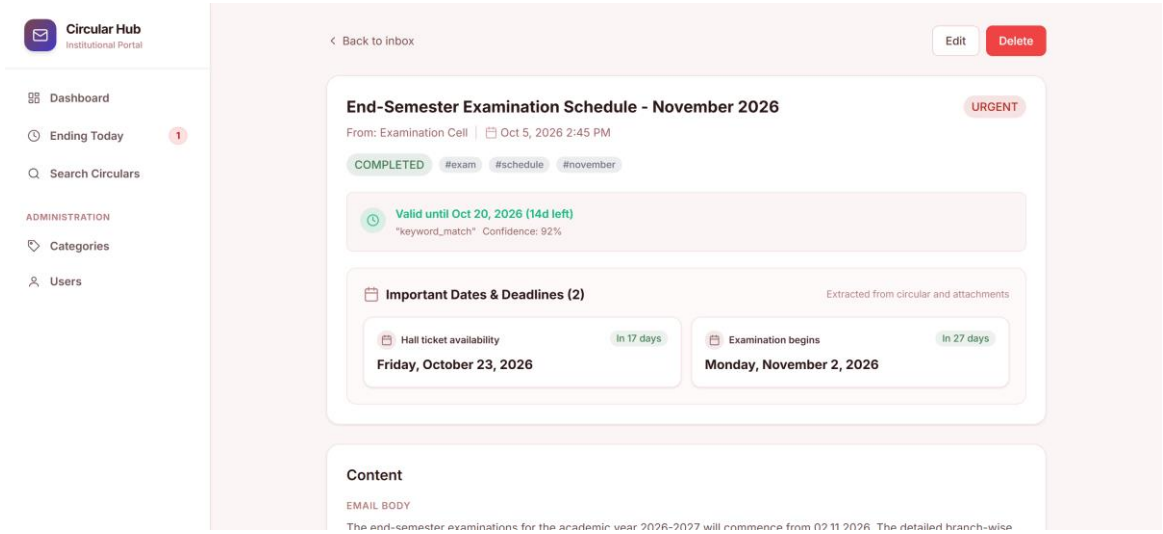
LOGIN & DASHBOARD :

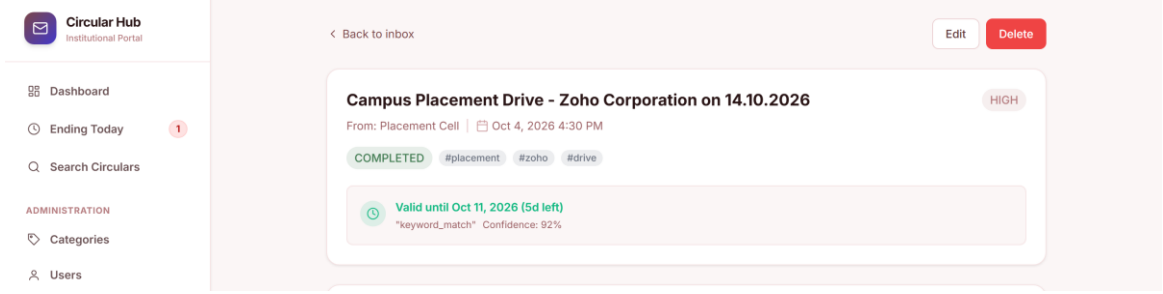




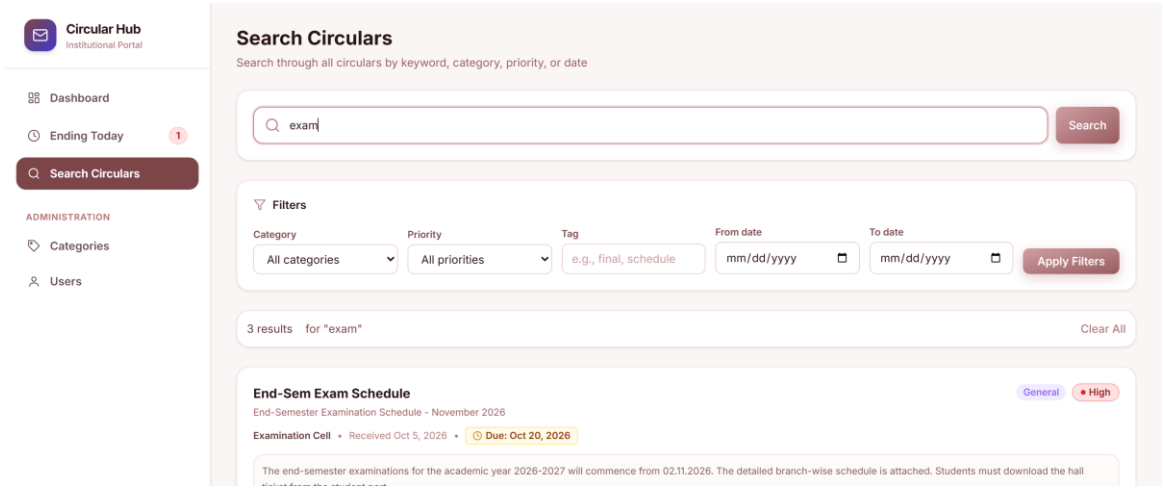

Circular Hub
Institutional Portal
Dashboard
Ending Today 1
Search Circulars
ADMINISTRATION
Categories
Users
Circular Hub
Centralized institutional circulars sorted by date and priority
Sync Mailbox
Search
Search keyword (instant)...
Filter by sender name
Enter name or email...
Category
All categories
Priority
All priorities
From date
mm/dd/yyyy
To date
mm/dd/yyyy
All circulars
MARK AS READ
CATEGORY
SENDER
TITLE
PRIORITY
RECEIVED DATE
DUE DATE
Read
General
Examination Cell
End-Sem Exam Schedule
End-Semester Examination...
High
Oct 5, 2026
Oct 20, 2026
Read
General
Accounts Office
Semester Fee Deadline
Semester Fee Payment - Last Date...
High
Oct 2, 2026
Oct 6, 2026
Read
General
Placement Cell
Zoho Placement Drive
Campus Placement Drive - Zoho...
High
Oct 4, 2026
Oct 11, 2026

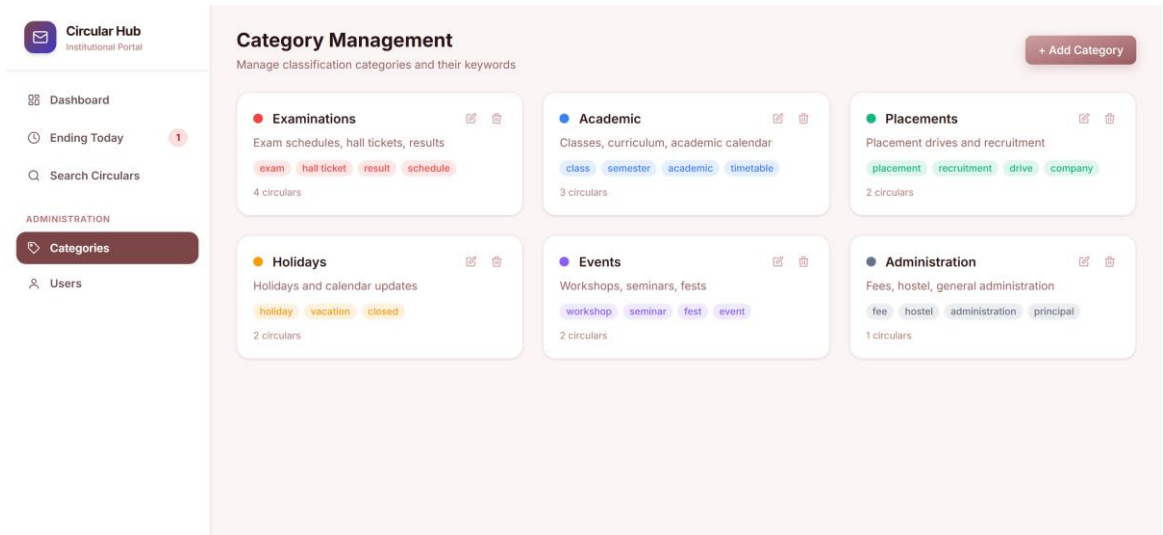

Circular Hub
Institutional Portal
Dashboard
Ending Today 1
Search Circulars
ADMINISTRATION
Categories
Users
Ending Today Circulars
All institutional circulars with due dates or deadlines today, sorted by priority
Sync Mailbox
Search
Search keyword (instant)...
Filter by sender name
Enter name or email...
Category
All categories
Priority
All priorities
From date
mm/dd/yyyy
To date
mm/dd/yyyy
All circulars
MARK AS READ
CATEGORY
SENDER
TITLE
PRIORITY
RECEIVED DATE
DUE DATE
Read
General
Accounts Office
Semester Fee Deadline
Semester Fee Payment - Last Date...
High
Oct 2, 2026
Oct 6, 2026

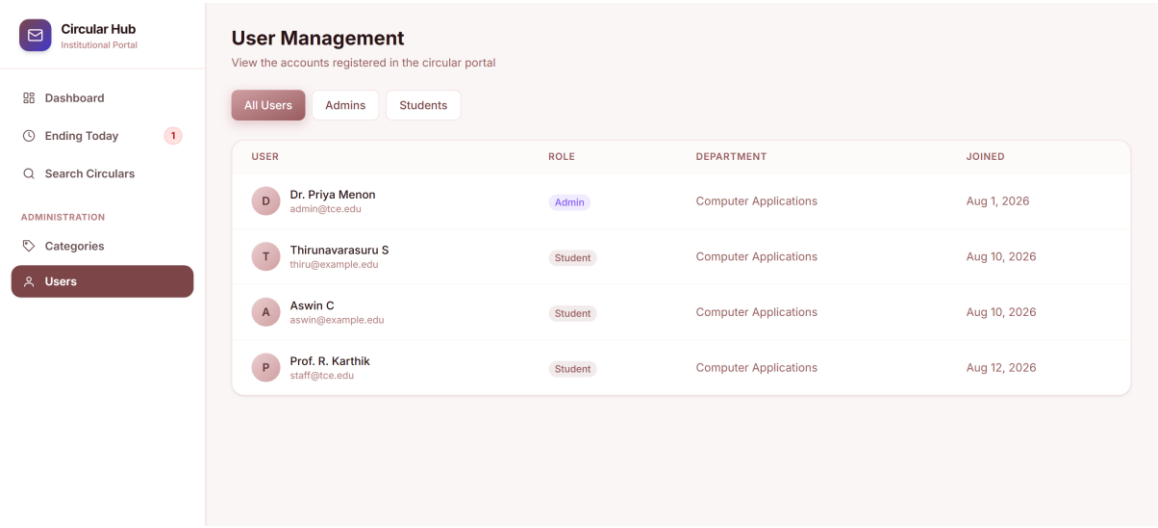

Circular Hub
Institutional Portal
Dashboard
Ending Today 1
Search Circulars
ADMINISTRATION
Categories
Users
Back to inbox
Edit
Delete
End-Semester Examination Schedule - November 2026
URGENT
From: Examination Cell
Oct 5, 2026 2:45 PM
COMPLETED
#exam
#schedule
#november
Valid until Oct 20, 2026 (14d left)
"keyword_match" Confidence: 92%
Important Dates & Deadlines (2)
Extracted from circular and attachments
Hall ticket availability
In 17 days
Friday, October 23, 2026
Examination begins
In 27 days
Monday, November 2, 2026
Content
EMAIL BODY

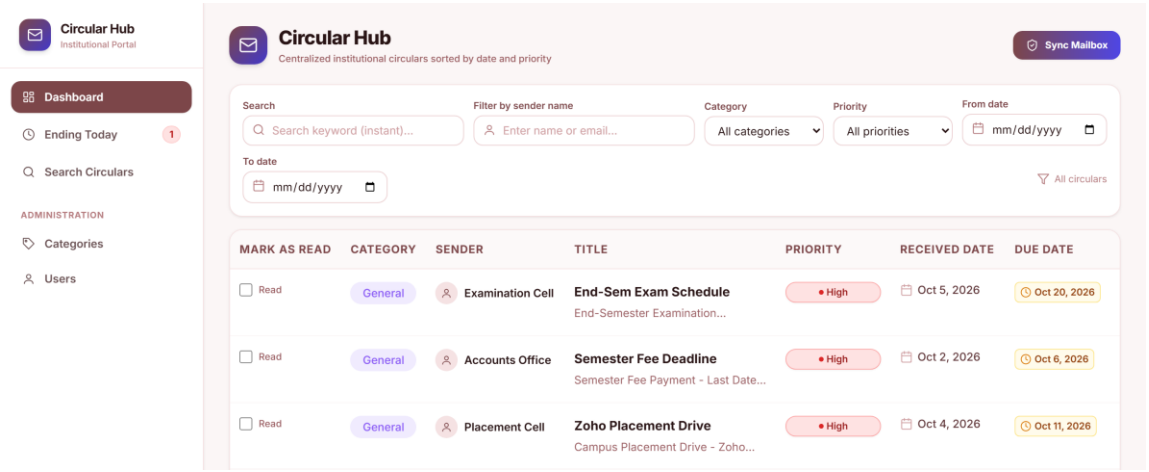

Circular Hub
Institutional Portal
Dashboard
Ending Today 1
Search Circulars
ADMINISTRATION
Categories
Users
Back to inbox
Edit
Delete
Campus Placement Drive - Zoho Corporation on 14.10.2026
HIGH
From: Placement Cell
Oct 4, 2026 4:30 PM
COMPLETED
#placement
#zoho
#drive
Valid until Oct 11, 2026 (5d left)
"keyword_match" Confidence: 92%

SEARCH & ADMIN :

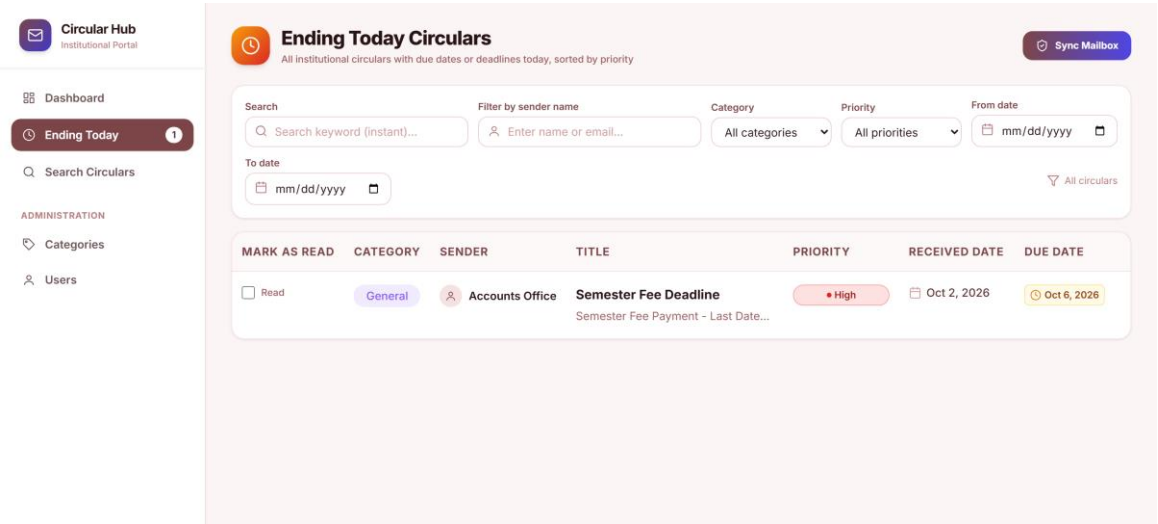


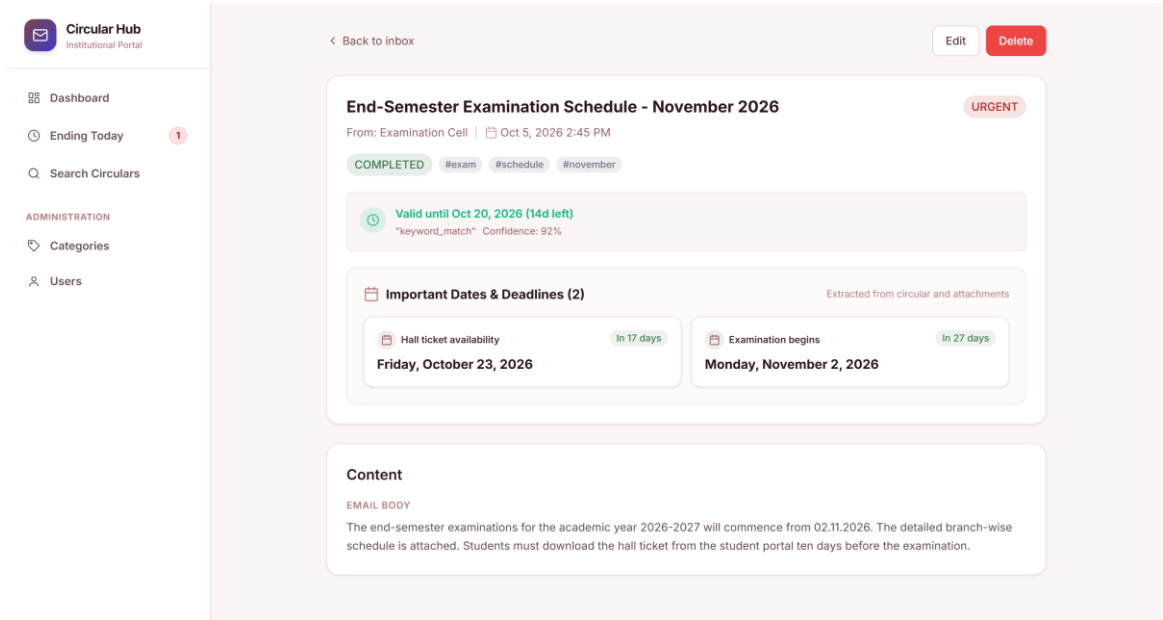


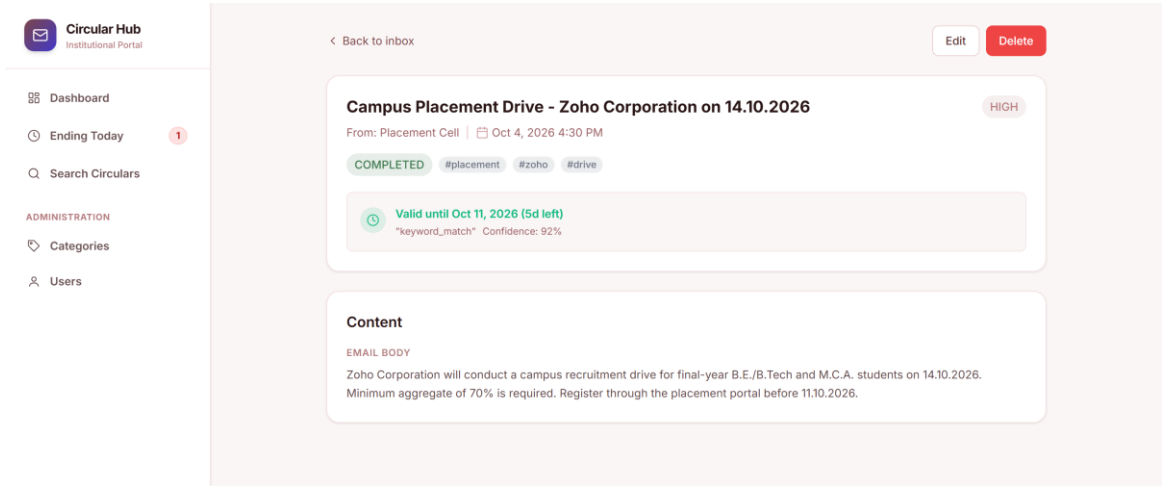


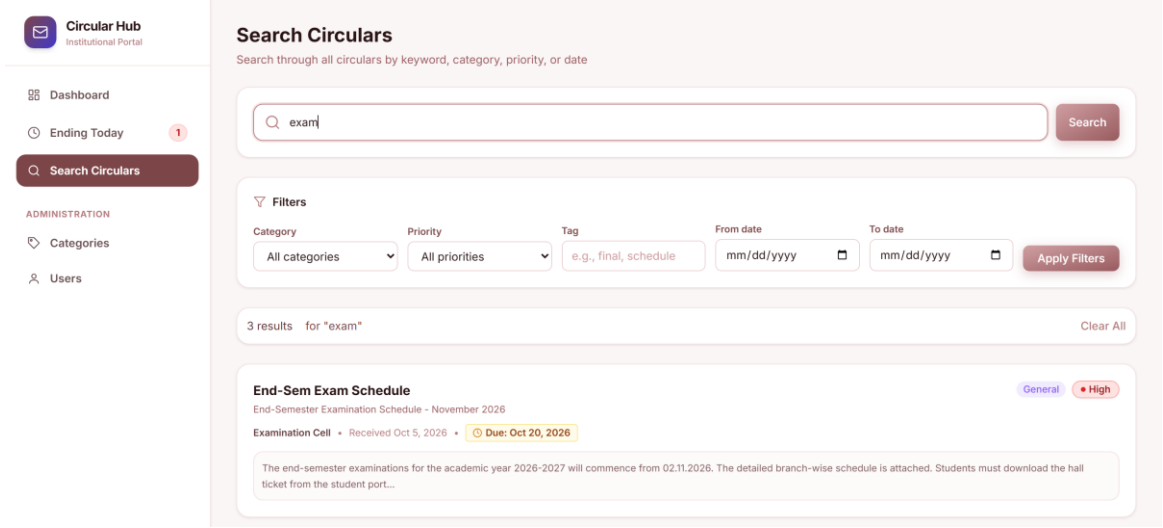


EXPIRY, DETAIL & SEARCH :

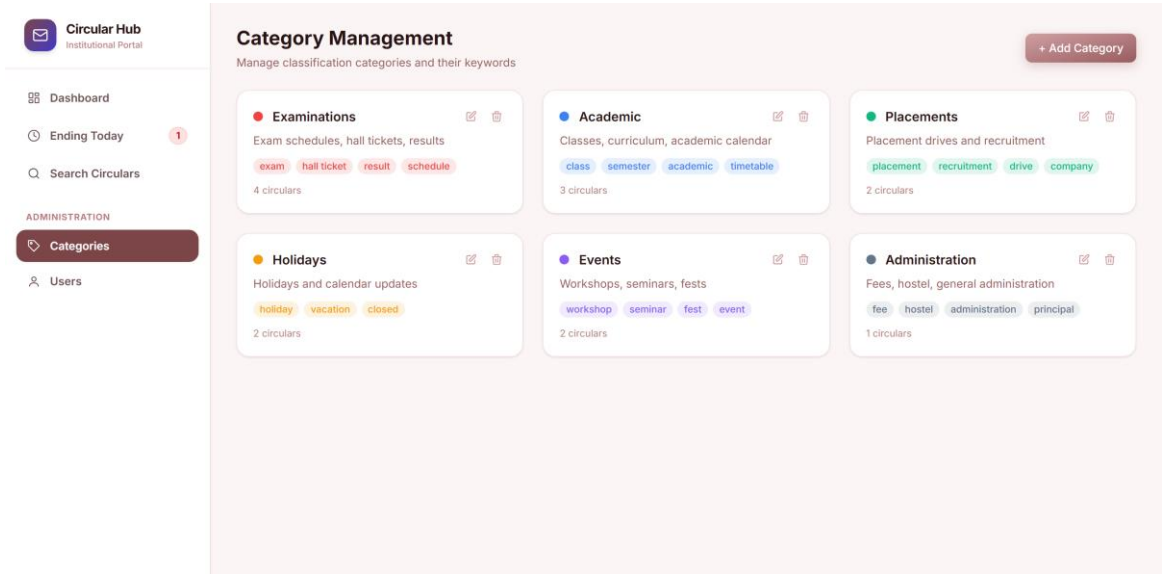


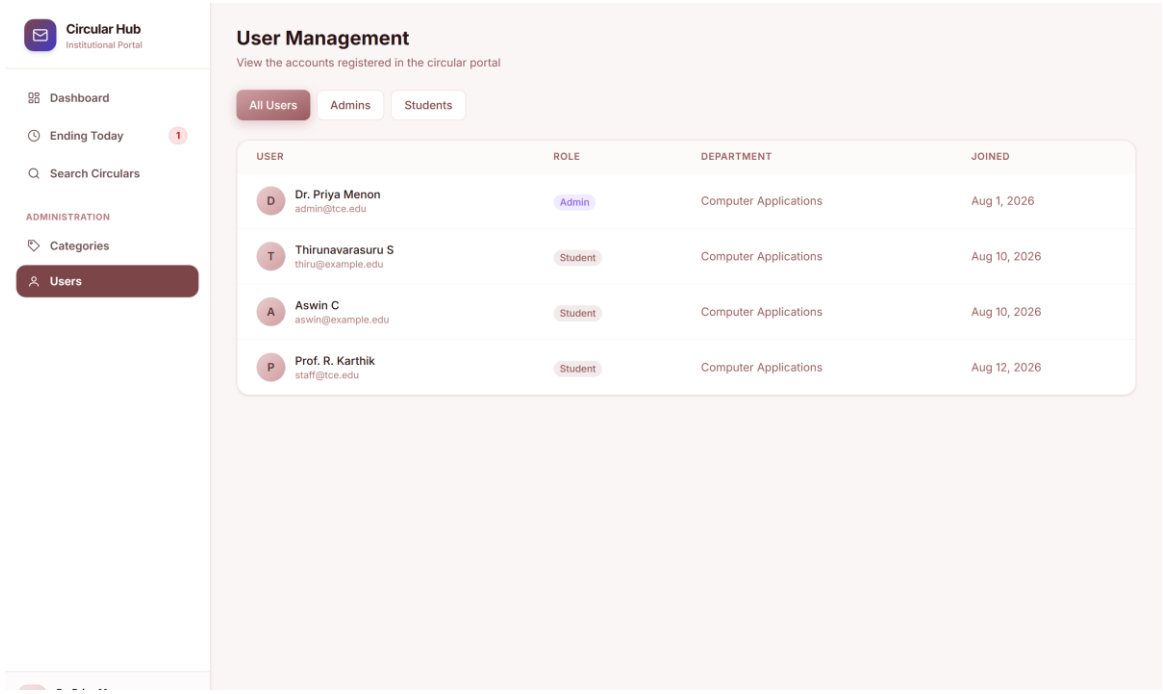

Circular Hub
Institutional Portal
Dashboard
Ending Today
1
Search Circulars
ADMINISTRATION
Categories
Users
Back to inbox
Edit
Delete
End-Semester Examination Schedule - November 2026
URGENT
From: Examination Cell
Oct 5, 2026 2:45 PM
COMPLETED
#exam
#schedule
#november
Valid until Oct 20, 2026 (14d left)
"keyword_match" Confidence: 92%
Important Dates & Deadlines (2)
Extracted from circular and attachments
Hall ticket availability
In 17 days
Friday, October 23, 2026
Examination begins
In 27 days
Monday, November 2, 2026
Content
EMAIL BODY
The end-semester examinations for the academic year 2026-2027 will commence from 02.11.2026. The detailed branch-wise schedule is attached. Students must download the hall ticket from the student portal ten days before the examination.

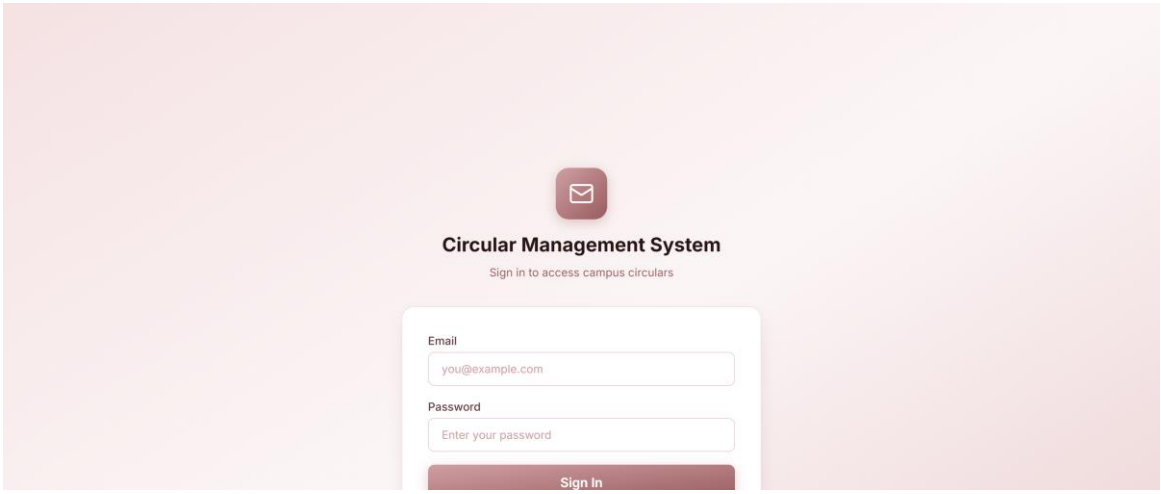

Circular Hub
Institutional Portal
Dashboard
Ending Today
1
Search Circulars
ADMINISTRATION
Categories
Users
Back to inbox
Edit
Delete
Campus Placement Drive - Zoho Corporation on 14.10.2026
HIGH
From: Placement Cell
Oct 4, 2026 4:30 PM
COMPLETED
#placement
#zoho
#drive
Valid until Oct 11, 2026 (5d left)
"keyword_match" Confidence: 92%
Content
EMAIL BODY
Zoho Corporation will conduct a campus recruitment drive for final-year B.E./B.Tech and M.C.A. students on 14.10.2026. Minimum aggregate of 70% is required. Register through the placement portal before 11.10.2026.


Circular Hub
Institutional Portal
Dashboard
Ending Today
1
Search Circulars
ADMINISTRATION
Categories
Users
Search Circulars
Search through all circulars by keyword, category, priority, or date
exam
Search
Filters
Category
All categories
Priority
All priorities
Tag
e.g., final, schedule
From date
mm/dd/yyyy
To date
mm/dd/yyyy
Apply Filters
3 results for "exam"
Clear All
End-Sem Exam Schedule
General
High
End-Semester Examination Schedule - November 2026
Examination Cell • Received Oct 5, 2026 • Due: Oct 20, 2026
The end-semester examinations for the academic year 2026-2027 will commence from 02.11.2026. The detailed branch-wise schedule is attached. Students must download the hall ticket from the student port...

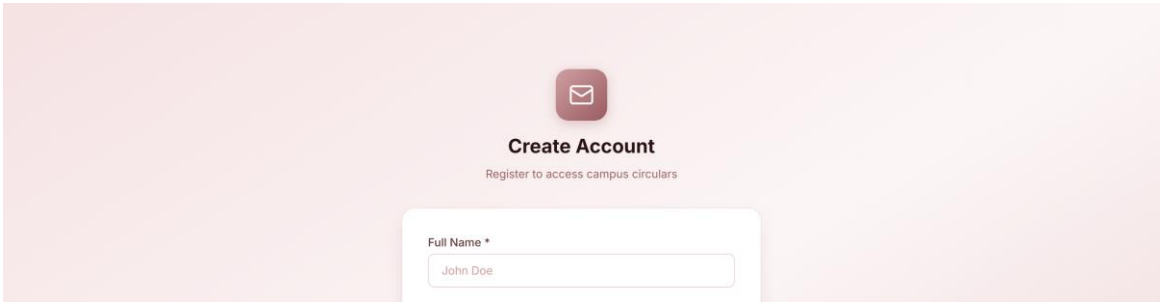
ADMIN & LOGIN :


Circular Hub
Institutional Portal
Dashboard
Ending Today
1
Search Circulars
ADMINISTRATION
Categories
Users
Category Management
Manage classification categories and their keywords
+ Add Category
Examinations
Exam schedules, hall tickets, results
exam
hall ticket
result
schedule
4 circulars
Academic
Classes, curriculum, academic calendar
class
semester
academic
timetable
3 circulars
Placements
Placement drives and recruitment
placement
recruitment
drive
company
2 circulars
Holidays
Holidays and calendar updates
holiday
vacation
closed
2 circulars
Events
Workshops, seminars, fests
workshop
seminar
fest
event
2 circulars
Administration
Fees, hostel, general administration
fee
hostel
administration
principal
1 circulars

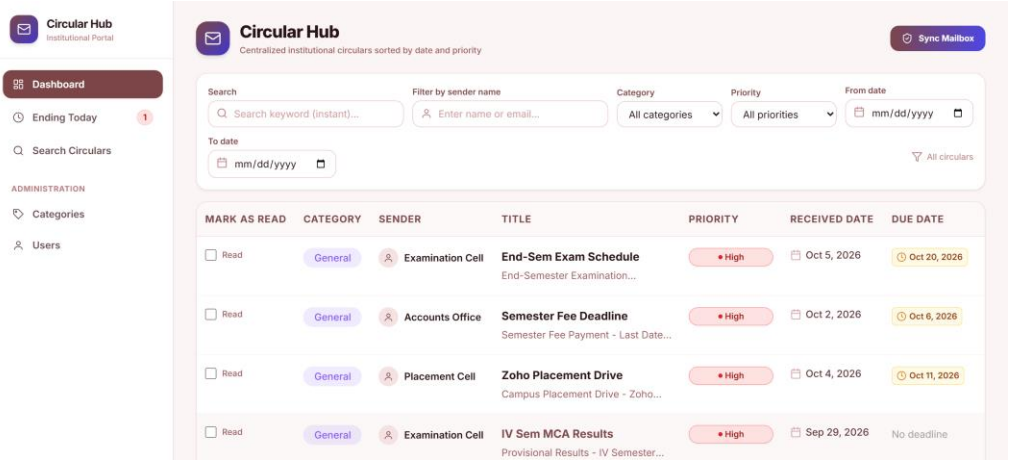

Circular Hub
Institutional Portal
Dashboard
Ending Today
1
Search Circulars
ADMINISTRATION
Categories
Users
User Management
View the accounts registered in the circular portal
All Users
Admins
Students
USER
ROLE
DEPARTMENT
JOINED
Dr. Priya Menon
admin@tce.edu
Admin
Computer Applications
Aug 1, 2026
Thirunavarasuru S
thiru@example.edu
Student
Computer Applications
Aug 10, 2026
Aswin C
aswin@example.edu
Student
Computer Applications
Aug 10, 2026
Prof. R. Karthik
staff@tce.edu
Student
Computer Applications
Aug 12, 2026

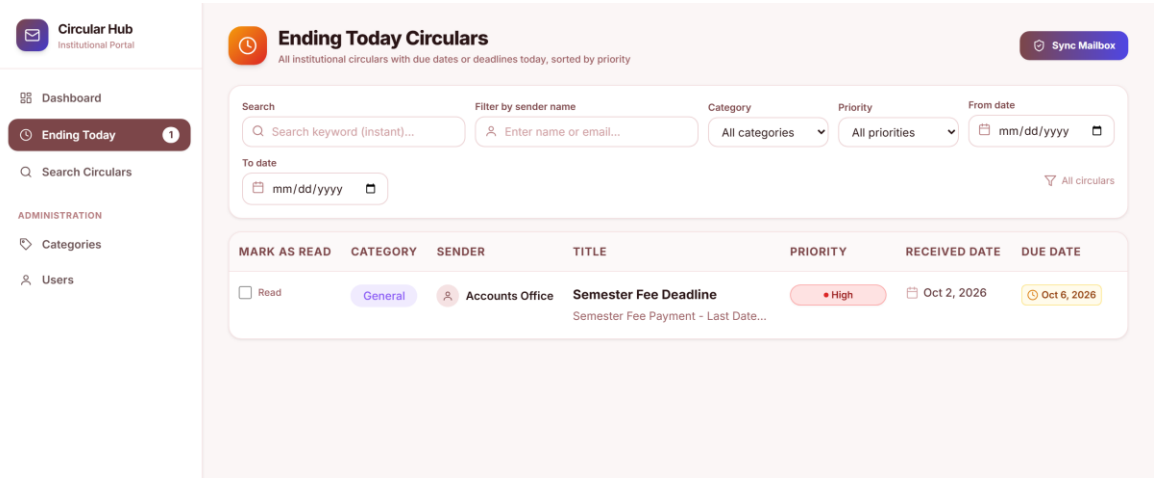


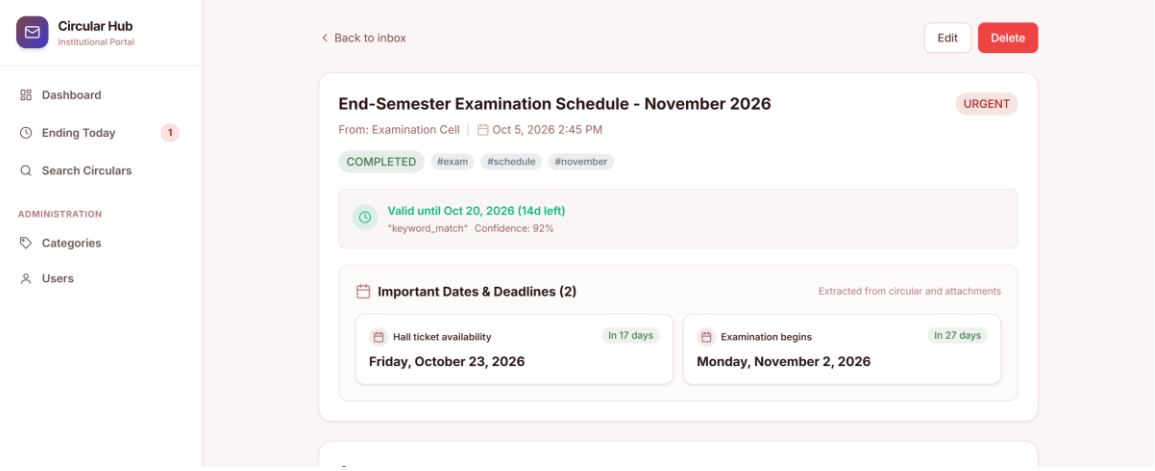
REGISTRATION :

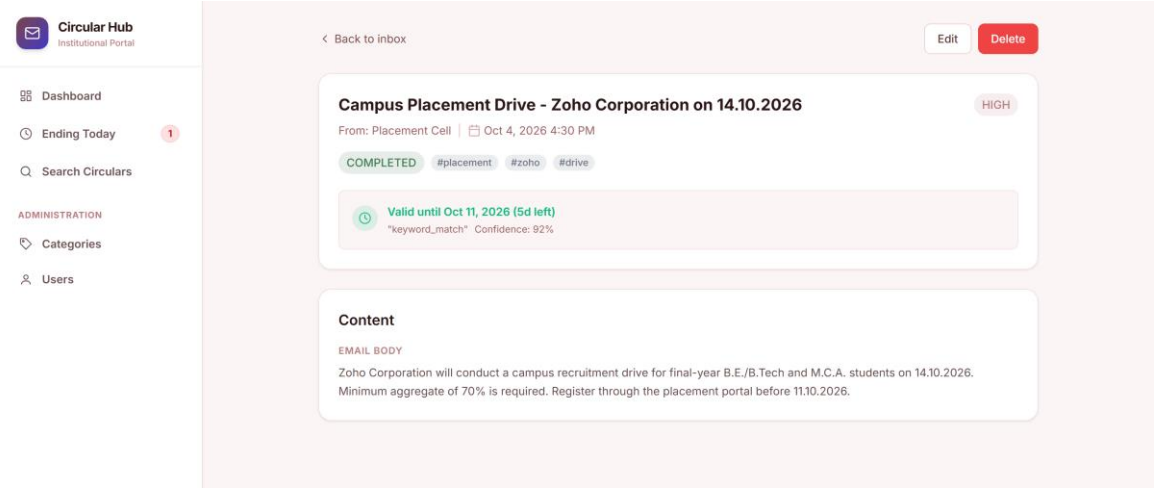


DASHBOARD & DETAIL :




Circular Hub
Institutional Portal
Dashboard
Ending Today 1
Search Circulars
ADMINISTRATION
Categories
Users
Ending Today Circulars
All institutional circulars with due dates or deadlines today, sorted by priority
Sync Mailbox
Search
Search keyword (instant)...
Filter by sender name
Enter name or email...
Category
All categories
Priority
All priorities
From date
mm/dd/yyyy
To date
mm/dd/yyyy
All circulars
MARK AS READ
CATEGORY
SENDER
TITLE
PRIORITY
RECEIVED DATE
DUE DATE
Read
General
Accounts Office
Semester Fee Deadline
Semester Fee Payment - Last Date...
High
Oct 2, 2026
Oct 6, 2026


Circular Hub
Institutional Portal
Dashboard
Ending Today 1
Search Circulars
ADMINISTRATION
Categories
Users
Back to inbox
Edit
Delete
End-Semester Examination Schedule - November 2026
URGENT
From: Examination Cell
Oct 5, 2026 2:45 PM
COMPLETED
#exam
#schedule
#november
Valid until Oct 20, 2026 (14d left)
"keyword_match" Confidence: 92%
Important Dates & Deadlines (2)
Extracted from circular and attachments
Hall ticket availability
In 17 days
Friday, October 23, 2026
Examination begins
In 27 days
Monday, November 2, 2026


Circular Hub
Institutional Portal
Dashboard
Ending Today 1
Search Circulars
ADMINISTRATION
Categories
Users
Back to inbox
Edit
Delete
Campus Placement Drive - Zoho Corporation on 14.10.2026
HIGH
From: Placement Cell
Oct 4, 2026 4:30 PM
COMPLETED
#placement
#zoho
#drive
Valid until Oct 11, 2026 (5d left)
"keyword_match" Confidence: 92%
Content
EMAIL BODY
Zoho Corporation will conduct a campus recruitment drive for final-year B.E./B.Tech and M.C.A. students on 14.10.2026.
Minimum aggregate of 70% is required. Register through the placement portal before 11.10.2026.